

Bd. 17 - Ganze Zahlen

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Kapitel 1

Einleitung

Die ganzen Zahlen entstehen aus Paaren natürlicher Zahlen. Das Paar (a, b) wird als formale Differenz $a - b$ gelesen. Da verschiedene Paare dieselbe Differenz beschreiben können, werden genau diejenigen Paare identifiziert, deren Kreuzsummen übereinstimmen. Die allgemeine Theorie der Äquivalenzrelationen und Quotienten aus Band 11 liefert anschließend die Menge der ganzen Zahlen.

Nach der Konstruktion werden Nachfolger, Vorgänger und die zweiseitige Induktion zunächst im Quotientenmodell bewiesen. Erst danach werden diese Eigenschaften als erweiterte Peano-Axiome registriert. Addition, Multiplikation und Negation werden stets erst nach einem Beweis ihrer Repräsentantenunabhängigkeit eingeführt.

Kapitel 2

Konstruktion der ganzen Zahlen

2.1 Differenzenpaare

Definition 17.2.1.1 (Träger der Differenzenpaare).

Mengen: $D_{\mathbb{Z}}$
Neue Symbole: $D_{\mathbb{Z}}$

$$D_{\mathbb{Z}} := \mathbb{N} \times \mathbb{N}$$

Definition 17.2.1.2 (Äquivalenz von Differenzenpaaren).

Mengen: a, b, c, d
Relationsoperatoren: $\sim_{\mathbb{Z}}$
Trägermenge: $D_{\mathbb{Z}}$
Neue Symbole: $\sim_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash (a, b) \sim_{\mathbb{Z}} (c, d) := a + d = b + c$$

Theorem 17.2.1.1 (Charakterisierung des Äquivalenzoperators auf Differenzenpaaren).

Mengen: a, b, c, d
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash (a, b) \sim_{\mathbb{Z}} (c, d) \leftrightarrow a + d = b + c$$

$$\begin{array}{l} 1 \quad 1 \quad a, b, c, d \in \mathbb{N} \quad A \\ 1 \quad 2 \quad (a, b) \sim_{\mathbb{Z}} (c, d) \leftrightarrow a + d = b + c \quad \text{Def. 17.2.1.2(1)} \end{array}$$

Theorem 17.2.1.2 (Reflexivität auf Differenzenpaaren).

Mengen: a, b
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$a, b \in \mathbb{N} \vdash (a, b) \sim_{\mathbb{Z}} (a, b)$$

$$\begin{array}{ll} 1 & 1 \ a, b \in \mathbb{N} \quad A \\ 1 & 2 \ a + b = b + a \quad \text{Th. 10.4.4.8(1)} \\ 1 & 3 \ (a, b) \sim_{\mathbb{Z}} (a, b) \quad \text{Th. 17.2.1.1(1,2)} \end{array}$$

Theorem 17.2.1.3 (Symmetrie auf Differenzenpaaren).

Mengen: a, b, c, d
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N}, (a, b) \sim_{\mathbb{Z}} (c, d) \vdash (c, d) \sim_{\mathbb{Z}} (a, b)$$

$$\begin{array}{ll} 1 & 1 \ a, b, c, d \in \mathbb{N} \quad A \\ 2 & 2 \ (a, b) \sim_{\mathbb{Z}} (c, d) \quad A \\ 1,2 & 3 \ a + d = b + c \quad \text{Th. 17.2.1.1(1,2)} \\ 1,2 & 4 \ c + b = b + c \quad \text{Th. 10.4.4.8(1)} \\ 1,2 & 5 \ b + c = a + d \quad \text{Th. 2.9.1.1(3)} \\ 1,2 & 6 \ a + d = d + a \quad \text{Th. 10.4.4.8(1)} \\ 1,2 & 7 \ c + b = d + a = E(4, 5, 6) \\ 1,2 & 8 \ (c, d) \sim_{\mathbb{Z}} (a, b) \quad \text{Th. 17.2.1.1(1,7)} \end{array}$$

Theorem 17.2.1.4 (Transitivität auf Differenzenpaaren).

Mengen: a, b, c, d, e, f
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$\begin{array}{l} a, b, c, d, e, f \in \mathbb{N}, (a, b) \sim_{\mathbb{Z}} (c, d), \\ (c, d) \sim_{\mathbb{Z}} (e, f) \vdash (a, b) \sim_{\mathbb{Z}} (e, f) \end{array}$$

Beweis.

Zusammenführen der Kreuzsummen

Th. 17.2.1.4(i)

$$\begin{array}{l} a, b, c, d, e, f \in \mathbb{N}, a + d = b + c, c + f = d + e \vdash \\ (a + f) + (c + d) = (b + e) + (c + d) \end{array}$$

1	$a, b, c, d, e, f \in \mathbb{N}$	A
2	$a + d = b + c$	A
3	$c + f = d + e$	A
1,2,3	$(a + d) + (c + f) = (b + c) + (d + e) = E(2, 3)$	
1	$(a + d) + (c + f) = (a + f) + (c + d)$	Th. 10.4.4.10(1)
1	$(b + c) + (d + e) = (b + e) + (c + d)$	Th. 10.4.4.10(1)
1,2,3	$(a + f) + (c + d) = (b + e) + (c + d) = E(4, 5, 6)$	

Kürzen der gemeinsamen Summe

Th. 17.2.1.4(ii)

$$a, b, c, d, e, f \in \mathbb{N}, (a + f) + (c + d) = (b + e) + (c + d)$$

$$\vdash a + f = b + e$$

1	$a, b, c, d, e, f \in \mathbb{N}$	A
2	$(a + f) + (c + d) = (b + e) + (c + d)$	A
1	$a + f \in \mathbb{N}$	Th. 10.4.4.1(1)
1	$b + e \in \mathbb{N}$	Th. 10.4.4.1(1)
1	$c + d \in \mathbb{N}$	Th. 10.4.4.1(1)
1,2	$a + f = b + e$	Th. 10.4.5.30(3,4,5,2)

Schluss:

1	$a, b, c, d, e, f \in \mathbb{N}$	A
2	$(a, b) \sim_{\mathbb{Z}} (c, d)$	A
3	$(c, d) \sim_{\mathbb{Z}} (e, f)$	A
1,2	$a + d = b + c$	Th. 17.2.1.1(1,2)
1,3	$c + f = d + e$	Th. 17.2.1.1(1,3)
1,2,3	$(a + f) + (c + d) = (b + e) + (c + d)$	Th. 17.2.1.4(i)(1,4,5)
1,2,3	$a + f = b + e$	Th. 17.2.1.4(ii)(1,6)
1,2,3	$(a, b) \sim_{\mathbb{Z}} (e, f)$	Th. 17.2.1.1(1,7)

□

Theorem 17.2.1.5 (Der Äquivalenzoperator auf Differenzenpaaren ist eine Äquivalenzrelation).

Mengen: $D_{\mathbb{Z}}$
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$\text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}})$

Beweis.

Reflexivität

Th. 17.2.1.5(i)

$$x \in D_{\mathbb{Z}} \vdash x \sim_{\mathbb{Z}} x$$

$$\begin{array}{ll}
1 & 1 \ x \in D_{\mathbb{Z}} \quad A \\
1 & 2 \ \exists a \in \mathbb{N} \exists b \in \mathbb{N} \ x = (a, b) \quad \text{Th. 3.16.5.4(1)} \\
3 & 3 \ a \in \mathbb{N} \wedge b \in \mathbb{N} \wedge x = (a, b) A \\
3 & 4 \ (a, b) \sim_{\mathbb{Z}} (a, b) \quad \text{Th. 17.2.1.2(3)} \\
3 & 5 \ x \sim_{\mathbb{Z}} x \quad = E(3, 4) \\
1 & 6 \ x \sim_{\mathbb{Z}} x \quad \exists E(2, 3, 5)
\end{array}$$

Symmetrie

Th. 17.2.1.5(ii)

$$x \sim_{\mathbb{Z}} y \vdash y \sim_{\mathbb{Z}} x$$

$$\begin{array}{ll}
1 & 1 \ x \sim_{\mathbb{Z}} y A \\
1 & 2 \ y \sim_{\mathbb{Z}} x \text{Th. 17.2.1.3(1)}
\end{array}$$

Transitivität

Th. 17.2.1.5(iii)

$$x \sim_{\mathbb{Z}} y, \ y \sim_{\mathbb{Z}} z \vdash x \sim_{\mathbb{Z}} z$$

$$\begin{array}{ll}
1 & 1 \ x \sim_{\mathbb{Z}} y A \\
2 & 2 \ y \sim_{\mathbb{Z}} z A \\
1,2 & 3 \ x \sim_{\mathbb{Z}} z \text{Th. 17.2.1.4(1,2)}
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 & x \in D_{\mathbb{Z}} \rightarrow x \sim_{\mathbb{Z}} x \text{Th. 17.2.1.5(i)} \\
2 & x \sim_{\mathbb{Z}} y \rightarrow y \sim_{\mathbb{Z}} x \text{Th. 17.2.1.5(ii)} \\
3 & 3 \ x \sim_{\mathbb{Z}} y \quad A \\
4 & 4 \ y \sim_{\mathbb{Z}} z \quad A \\
3,4 & 5 \ x \sim_{\mathbb{Z}} z \quad \text{Th. 17.2.1.5(iii)(3,4)} \\
6 & \text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}}) \quad \text{Def. 11.2.2.1(1,2,5)}
\end{array}$$

□

2.2 Quotient und Klassen

Definition 17.2.2.1 (Menge der ganzen Zahlen).

Mengen: $D_{\mathbb{Z}}, \mathbb{Z}$
Relationsoperatoren: $\sim_{\mathbb{Z}}$
Neue Symbole: $\mathbb{Z}$

$$\mathbb{Z} := D_{\mathbb{Z}} / \sim_{\mathbb{Z}}$$

Definition 17.2.2.2 (Klasse eines Differenzenpaares).

Mengen: $a, b, D_{\mathbb{Z}}$

Relationsoperatoren: $\sim_{\mathbb{Z}}$

Neue Symbole: $[a, b]_{\mathbb{Z}}$

$$a, b \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} := [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$$

Theorem 17.2.2.1 (Klassen sind ganze Zahlen).

Mengen: a, b

$$a, b \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} \in \mathbb{Z}$$

1	1	$a, b \in \mathbb{N}$	A
1	2	$(a, b) \in D_{\mathbb{Z}}$	Th. 3.16.5.5(1)
	3	$\text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}})$	Th. 17.2.1.5
1	4	$[(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} \in D_{\mathbb{Z}} / \sim_{\mathbb{Z}}$	Th. 11.4.2.2(3, 2)
1	5	$[a, b]_{\mathbb{Z}} = [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$	Def. 17.2.2.2(1)
	6	$\mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}}$	Def. 17.2.2.1
1	7	$[a, b]_{\mathbb{Z}} \in \mathbb{Z}$	$= E(5, 6, 4)$

Theorem 17.2.2.2 (Gleichheit von Ganzzahlklassen).

Mengen: a, b, c, d

Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow a + d = b + c$$

Beweis.

Klassenkriterium

Th. 17.2.2.2(i)

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow (a, b) \sim_{\mathbb{Z}} (c, d)$$

1	1	$a, b, c, d \in \mathbb{N}$	A
	2	$\text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}})$	Th. 17.2.1.5
1	3	$(a, b) \in D_{\mathbb{Z}}$	Th. 3.16.5.5(1)
1	4	$(c, d) \in D_{\mathbb{Z}}$	Th. 3.16.5.5(1)
1	5	$[(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} = [(c, d)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} \leftrightarrow (a, b) \sim_{\mathbb{Z}} (c, d)$	Th. 11.4.1.7(2, 3, 4)
1	6	$[a, b]_{\mathbb{Z}} = [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$	Def. 17.2.2.2(1)
1	7	$[c, d]_{\mathbb{Z}} = [(c, d)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$	Def. 17.2.2.2(1)

$$1 \quad 8 \quad [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow (a, b) \sim_{\mathbb{Z}} (c, d) \quad = E(5, 6, 7)$$

Schluss:

$$\begin{array}{ll} 1 \quad 1 \quad a, b, c, d \in \mathbb{N} & A \\ 1 \quad 2 \quad [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow (a, b) \sim_{\mathbb{Z}} (c, d) & \text{Th. 17.2.2.2(i)(1)} \\ 1 \quad 3 \quad (a, b) \sim_{\mathbb{Z}} (c, d) \leftrightarrow a + d = b + c & \text{Th. 17.2.1.1(1)} \\ 1 \quad 4 \quad [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow a + d = b + c & \text{Th. 2.2.3.5(2,3)} \end{array}$$

□

Theorem 17.2.2.3 (Jede ganze Zahl besitzt ein Differenzenpaar).

Mengen: $z, a, b, D_{\mathbb{Z}}$

$$z \in \mathbb{Z} \vdash \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$$

Beweis.

Quotientenrepräsentant

Th. 17.2.2.3(i)

$$z \in \mathbb{Z} \vdash \exists x \in D_{\mathbb{Z}} z = [x]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$$

$$\begin{array}{ll} 1 \quad 1 \quad z \in \mathbb{Z} & A \\ 2 \quad \mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}} & \text{Def. 17.2.2.1} \\ 1 \quad 3 \quad z \in D_{\mathbb{Z}} / \sim_{\mathbb{Z}} & = E(2, 1) \\ 4 \quad \text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}}) & \text{Th. 17.2.1.5} \\ 1 \quad 5 \quad \exists x \in D_{\mathbb{Z}} z = [x]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & \text{Th. 11.4.2.4(4,3)} \end{array}$$

Schluss:

$$\begin{array}{ll} 1 \quad 1 \quad z \in \mathbb{Z} & A \\ 1 \quad 2 \quad \exists x \in D_{\mathbb{Z}} z = [x]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & \text{Th. 17.2.2.3(i)(1)} \\ 3 \quad 3 \quad x \in D_{\mathbb{Z}} \wedge z = [x]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & A \\ 3 \quad 4 \quad \exists a \in \mathbb{N} \exists b \in \mathbb{N} x = (a, b) & \text{Th. 3.16.5.4}(\wedge E1(3)) \\ 5 \quad 5 \quad a, b \in \mathbb{N} \wedge x = (a, b) & A \\ 3,5 \quad 6 \quad z = [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & = E(5, \wedge E2(3)) \\ 5 \quad 7 \quad [a, b]_{\mathbb{Z}} = [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & \text{Def. 17.2.2.2(5)} \\ 3,5 \quad 8 \quad z = [a, b]_{\mathbb{Z}} & \text{Th. 2.9.1.3(6,7)} \\ 3 \quad 9 \quad \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} & \exists E(4, 5, 8) \\ 1 \quad 10 \quad \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} & \exists E(2, 3, 9) \end{array}$$

□

2.3 Einbettung der natürlichen Zahlen

Definition 17.2.3.1 (Natürliche Einbettung).

Mengen: n
Funktionensymbol: $\iota_{\mathbb{Z}}$
Neue Symbole: $\iota_{\mathbb{Z}}$

$$\iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z},$$

$$\forall n \in \mathbb{N} (\iota_{\mathbb{Z}}(n) := [n, 0]_{\mathbb{Z}})$$

- | | | | |
|---|---|---|----------------------------------|
| 1 | 1 | $n \in \mathbb{N}$ | A |
| | 2 | $0 \in \mathbb{N}$ | $Ax. 10.3.1$ |
| 1 | 3 | $[n, 0]_{\mathbb{Z}} \in \mathbb{Z}$ | $Th. 17.2.2.1(1, 2)$ |
| | 4 | $\forall n \in \mathbb{N} ([n, 0]_{\mathbb{Z}} \in \mathbb{Z})$ | $\forall I(\rightarrow I(1, 3))$ |

Theorem 17.2.3.1 (Typisierung der natürlichen Einbettung).

Funktion: $\iota_{\mathbb{Z}}$

$$\iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z}$$

- 1 $\iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z}$ *Def. 17.2.3.1*

Theorem 17.2.3.2 (Grundgleichung der natürlichen Einbettung).

Mengen: n
Funktion: $\iota_{\mathbb{Z}}$

$$n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}$$

- | | | | |
|---|---|--|----------------------------------|
| 1 | 1 | $n \in \mathbb{N}$ | A |
| | 2 | $\forall u \in \mathbb{N} (\iota_{\mathbb{Z}}(u) = [u, 0]_{\mathbb{Z}})$ | <i>Def. 17.2.3.1</i> |
| 1 | 3 | $\iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}$ | $\rightarrow E(\forall E(2), 1)$ |

Theorem 17.2.3.3 (Injektivität der natürlichen Einbettung).

Mengen: m, n
Funktion: $\iota_{\mathbb{Z}}$

$$\iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z}$$

Beweis.

Gleichheit von eingebetteten Zahlen

Th. 17.2.3.3(i)

$$m, n \in \mathbb{N}, \iota_{\mathbb{Z}}(m) = \iota_{\mathbb{Z}}(n) \vdash m = n$$

$$\begin{array}{ll} 1 & m, n \in \mathbb{N} \quad A \\ 2 & \iota_{\mathbb{Z}}(m) = \iota_{\mathbb{Z}}(n) \quad A \\ 1 & \iota_{\mathbb{Z}}(m) = [m, 0]_{\mathbb{Z}} \quad \text{Th. 17.2.3.2(1)} \\ 1 & \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}} \quad \text{Th. 17.2.3.2(1)} \\ 1,2 & [m, 0]_{\mathbb{Z}} = [n, 0]_{\mathbb{Z}} = E(3, 4, 2) \\ 1,2 & m + 0 = 0 + n \quad \text{Th. 17.2.2.2(1,5)} \\ 1 & m + 0 = m \quad \text{Th. 10.4.4.2(1)} \\ 1 & 0 + n = n \quad \text{Th. 10.4.4.4(1)} \\ 1,2 & m = n \quad = E(7, 8, 6) \end{array}$$

Schluss:

$$\begin{array}{ll} 1 & \iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z} \quad \text{Th. 17.2.3.1} \\ 2 & m \in \mathbb{N} \quad A \\ 3 & n \in \mathbb{N} \quad A \\ 4 & \iota_{\mathbb{Z}}(m) = \iota_{\mathbb{Z}}(n) \quad A \\ 2,3,4 & m = n \quad \text{Th. 17.2.3.3(i)(2,3,4)} \\ 6 & \iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z} \quad \text{Def. 6.2.1.1(1, } \forall I(\rightarrow I(2, 3, 4, 5))) \end{array}$$

□

Definition 17.2.3.2 (Null der ganzen Zahlen).

Mengen: $0_{\mathbb{Z}}$
Neue Symbole: $0_{\mathbb{Z}}$

$$0_{\mathbb{Z}} := \iota_{\mathbb{Z}}(0)$$

Definition 17.2.3.3 (Eins der ganzen Zahlen).

Mengen: $1_{\mathbb{Z}}$
Neue Symbole: $1_{\mathbb{Z}}$

$$1_{\mathbb{Z}} := \iota_{\mathbb{Z}}(1)$$

Theorem 17.2.3.4 (Nullklasse).

Mengen: $0_{\mathbb{Z}}$

$$0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}$$

- 1 $0_{\mathbb{Z}} = \iota_{\mathbb{Z}}(0)$ *Def.* 17.2.3.2
- 2 $0 \in \mathbb{N}$ *Ax.* 10.3.1
- 3 $\iota_{\mathbb{Z}}(0) = [0, 0]_{\mathbb{Z}}$ *Th.* 17.2.3.2(2)
- 4 $0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}$ *Th.* 2.9.1.2(1, 3)

Theorem 17.2.3.5 (Die Ganzzahlnull liegt im Quotienten).

Mengen: $0_{\mathbb{Z}}$

$$0_{\mathbb{Z}} \in \mathbb{Z}$$

- 1 $0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}$ *Th.* 17.2.3.4
- 2 $0 \in \mathbb{N}$ *Ax.* 10.3.1
- 3 $[0, 0]_{\mathbb{Z}} \in \mathbb{Z}$ *Th.* 17.2.2.1(2, 2)
- 4 $0_{\mathbb{Z}} \in \mathbb{Z}$ $= E(1, 3)$

Theorem 17.2.3.6 (Die Ganzzahleins liegt im Quotienten).

Mengen: $1_{\mathbb{Z}}$

$$1_{\mathbb{Z}} \in \mathbb{Z}$$

- 1 $1_{\mathbb{Z}} = \iota_{\mathbb{Z}}(1)$ *Def.* 17.2.3.3
- 2 $1 \in \mathbb{N}$ *Th.* 10.4.4.11
- 3 $\iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z}$ *Th.* 17.2.3.1
- 4 $\iota_{\mathbb{Z}}(1) \in \mathbb{Z}$ *Th.* 5.2.3.8(3, 2)
- 5 $1_{\mathbb{Z}} \in \mathbb{Z}$ $= E(1, 4)$

2.4 Negation und Addition

2.4.1 Negation

Theorem 17.2.4.1 (Repräsentantenunabhängigkeit der Negation).

Mengen: a, b, c, d

$$a, b, c, d \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \vdash \\ [b, a]_{\mathbb{Z}} = [d, c]_{\mathbb{Z}}$$

Beweis.

Kreuzsummen des Ausgangspaares

Th. 17.2.4.1(i)

$$a, b, c, d \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \vdash a + d = b + c$$

$$\begin{array}{ll} 1 & 1 \ a, b, c, d \in \mathbb{N} \quad A \\ 2 & 2 \ [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} A \\ 1,2 & 3 \ a + d = b + c \quad \text{Th. 17.2.2.2(1,2)} \end{array}$$

Vertauschen der Koordinaten:

$$\begin{array}{ll} 1 & 1 \ a, b, c, d \in \mathbb{N} \quad A \\ 2 & 2 \ [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} A \\ 1,2 & 3 \ a + d = b + c \quad \text{Th. 17.2.4.1(i)(1,2)} \\ 1,2 & 4 \ b + c = a + d \quad \text{Th. 2.9.1.1(3)} \\ 1,2 & 5 \ [b, a]_{\mathbb{Z}} = [d, c]_{\mathbb{Z}} \text{Th. 17.2.2.2(1,4)} \end{array}$$

□

Theorem 17.2.4.2 (Quotientenabstieg der Negation).

Mengen: $z, u, v, a, b, c, d, p, q$

$$z \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}})$$

Beweis.

Existenz aus einem Repräsentanten

Th. 17.2.4.2(i)

$$z \in \mathbb{Z} \vdash \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}})$$

$$\begin{array}{ll} 1 & 1 \ z \in \mathbb{Z} \quad A \\ 1 & 2 \ \exists a \in \mathbb{N} \exists b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}} \quad \text{Th. 17.2.2.3(1)} \\ 3 & 3 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \quad A \\ 3 & 4 \ [b, a]_{\mathbb{Z}} \in \mathbb{Z} \quad \text{Th. 17.2.2.1(3)} \\ & 5 \ [b, a]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}} \quad = I \\ 3 & 6 \ \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge [b, a]_{\mathbb{Z}} = [q, p]_{\mathbb{Z}}) \quad \exists I(\wedge I(3, 5)) \\ 3 & 7 \ [b, a]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge [b, a]_{\mathbb{Z}} = [q, p]_{\mathbb{Z}}) \wedge I(4, 6) \\ 3 & 8 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}}) \quad \exists I(7) \\ 1 & 9 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}}) \quad \exists E(2, 3, 8) \end{array}$$

Eindeutigkeit für feste Repräsentanten

Th. 17.2.4.2(ii)

$$a, b, c, d \in \mathbb{N}, \quad z = [a, b]_{\mathbb{Z}}, \quad u = [b, a]_{\mathbb{Z}}, \\ z = [c, d]_{\mathbb{Z}}, \quad v = [d, c]_{\mathbb{Z}} \vdash u = v$$

$$\begin{array}{ll} 1 & 1 \ a, b, c, d \in \mathbb{N} \quad A \\ 2 & 2 \ z = [a, b]_{\mathbb{Z}} \quad A \\ 3 & 3 \ u = [b, a]_{\mathbb{Z}} \quad A \\ 4 & 4 \ z = [c, d]_{\mathbb{Z}} \quad A \\ 5 & 5 \ v = [d, c]_{\mathbb{Z}} \quad A \\ 2,4 & 6 \ [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} = E(2, 4) \\ 1,2,4 & 7 \ [b, a]_{\mathbb{Z}} = [d, c]_{\mathbb{Z}} \text{ Th. 17.2.4.1(1,6)} \\ 1,2,3,4,5 & 8 \ u = v \quad = E(3, 5, 7) \end{array}$$

Eindeutigkeit beliebiger Ergebniszeugen

Th. 17.2.4.2(iii)

$$(u \in \mathbb{Z} \wedge \exists a \in \mathbb{N} \exists b \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge u = [b, a]_{\mathbb{Z}})), \\ (v \in \mathbb{Z} \wedge \exists c \in \mathbb{N} \exists d \in \mathbb{N} (z = [c, d]_{\mathbb{Z}} \wedge v = [d, c]_{\mathbb{Z}})) \vdash u = v$$

$$\begin{array}{ll} 1 & 1 \ u \in \mathbb{Z} \wedge \exists a \in \mathbb{N} \exists b \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge u = [b, a]_{\mathbb{Z}}) A \\ 2 & 2 \ v \in \mathbb{Z} \wedge \exists c \in \mathbb{N} \exists d \in \mathbb{N} (z = [c, d]_{\mathbb{Z}} \wedge v = [d, c]_{\mathbb{Z}}) A \\ 1 & 3 \ \exists a \in \mathbb{N} \exists b \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge u = [b, a]_{\mathbb{Z}}) \quad \wedge E2(1) \\ 2 & 4 \ \exists c \in \mathbb{N} \exists d \in \mathbb{N} (z = [c, d]_{\mathbb{Z}} \wedge v = [d, c]_{\mathbb{Z}}) \quad \wedge E2(2) \\ 5 & 5 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge u = [b, a]_{\mathbb{Z}} \quad A \\ 6 & 6 \ c, d \in \mathbb{N} \wedge z = [c, d]_{\mathbb{Z}} \wedge v = [d, c]_{\mathbb{Z}} \quad A \\ 5 & 7 \ a, b \in \mathbb{N} \quad \wedge E1(5) \\ 5 & 8 \ z = [a, b]_{\mathbb{Z}} \quad \wedge E1(\wedge E2(5)) \\ 5 & 9 \ u = [b, a]_{\mathbb{Z}} \quad \wedge E2(\wedge E2(5)) \\ 6 & 10 \ c, d \in \mathbb{N} \quad \wedge E1(6) \\ 6 & 11 \ z = [c, d]_{\mathbb{Z}} \quad \wedge E1(\wedge E2(6)) \\ 6 & 12 \ v = [d, c]_{\mathbb{Z}} \quad \wedge E2(\wedge E2(6)) \\ 5,6 & 13 \ a, b, c, d \in \mathbb{N} \quad \wedge I(7, 10) \\ 5,6 & 14 \ u = v \quad \text{Th. 17.2.4.2(ii)(13,8,9,11,12)} \\ 5 & 15 \ u = v \quad \exists E(4, 6, 14) \\ 1,2 & 16 \ u = v \quad \exists E(3, 5, 15) \end{array}$$

Abschluss des Abstiegs

Th. 17.2.4.2(iv)

$$z \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}})$$

$$\begin{array}{ll} 1 & 1 \ z \in \mathbb{Z} \quad A \\ 1 & 2 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}}) \text{ Th. 17.2.4.2(i)(1)} \\ 3 & 3 \ u \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}}) A \end{array}$$

$$\begin{array}{ll}
4 & 4 \ v \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge v = [q, p]_{\mathbb{Z}}) \ A \\
3,4 & 5 \ u = v \quad \text{Th. 17.2.4.2(iii)(3,4)} \\
1 & 6 \ \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}}) \ \exists! I(2, 3, 4, 5)
\end{array}$$

□

Der vorangehende Satz zeigt, dass die folgende Klassenregel unabhängig vom gewählten Differenzenpaar genau ein Ergebnis in $\mathbb{Z}$ festlegt.

Definition 17.2.4.1 (Negation ganzer Zahlen).

Mengen: a, b
Neue Symbole: $-$

$$a, b \in \mathbb{N} \vdash -[a, b]_{\mathbb{Z}} := [b, a]_{\mathbb{Z}}$$

Theorem 17.2.4.3 (Abgeschlossenheit der Negation).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash -z \in \mathbb{Z}$$

$$\begin{array}{ll}
1 & 1 \ z \in \mathbb{Z} \quad A \\
1 & 2 \ \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} \quad \text{Th. 17.2.2.3(1)} \\
3 & 3 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \quad A \\
3 & 4 \ -[a, b]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}} \quad \text{Def. 17.2.4.1(3)} \\
3 & 5 \ [b, a]_{\mathbb{Z}} \in \mathbb{Z} \quad \text{Th. 17.2.2.1(3)} \\
3 & 6 \ -z \in \mathbb{Z} \quad = E(3, 4, 5) \\
1 & 7 \ -z \in \mathbb{Z} \quad \exists E(2, 3, 6)
\end{array}$$

Theorem 17.2.4.4 (Doppelte Negation).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash -(-z) = z$$

$$\begin{array}{ll}
1 & 1 \ z \in \mathbb{Z} \quad A \\
1 & 2 \ \exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} \quad \text{Th. 17.2.2.3(1)} \\
3 & 3 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \quad A \\
3 & 4 \ -[a, b]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}} \quad \text{Def. 17.2.4.1(3)} \\
3 & 5 \ -[b, a]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} \quad \text{Def. 17.2.4.1(3)} \\
3 & 6 \ -(-z) = z \quad = E(3, 4, 5)
\end{array}$$

$$1 \quad 7 \quad -(-z) = z \quad \exists E(2, 3, 6)$$

2.4.2 Addition

Theorem 17.2.4.5 (Repräsentantenunabhängigkeit der Addition).

Mengen: $a, b, a', b', c, d, c', d'$

$$a, b, a', b', c, d, c', d' \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}, [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\ [a + c, b + d]_{\mathbb{Z}} = [a' + c', b' + d']_{\mathbb{Z}}$$

Beweis.

Kreuzsummen der beiden Eingaben

Th. 17.2.4.5(i)

$$a, b, a', b', c, d, c', d' \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}, [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\ a + b' = b + a' \quad \wedge \quad c + d' = d + c'$$

$$\begin{array}{ll} 1 \quad 1 \quad a, b, a', b', c, d, c', d' \in \mathbb{N} & A \\ 2 \quad 2 \quad [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}} & A \\ 3 \quad 3 \quad [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} & A \\ 1,2 \quad 4 \quad a + b' = b + a' & \text{Th. 17.2.2.2(1,2)} \\ 1,3 \quad 5 \quad c + d' = d + c' & \text{Th. 17.2.2.2(1,3)} \\ 1,2,3 \quad 6 \quad a + b' = b + a' \wedge c + d' = d + c' & \wedge I(4, 5) \end{array}$$

Zusammenführen und Umsortieren

Th. 17.2.4.5(ii)

$$a, b, a', b', c, d, c', d' \in \mathbb{N}, a + b' = b + a', c + d' = d + c' \vdash \\ (a + c) + (b' + d') = (b + d) + (a' + c')$$

$$\begin{array}{ll} 1 \quad 1 \quad a, b, a', b', c, d, c', d' \in \mathbb{N} & A \\ 2 \quad 2 \quad a + b' = b + a' & A \\ 3 \quad 3 \quad c + d' = d + c' & A \\ 1,2,3 \quad 4 \quad (a + b') + (c + d') = (b + a') + (d + c') = E(2, 3) \\ 1 \quad 5 \quad (a + b') + (c + d') = (a + c) + (b' + d') & \text{Th. 10.4.4.10(1)} \\ 1 \quad 6 \quad (b + a') + (d + c') = (b + d) + (a' + c') & \text{Th. 10.4.4.10(1)} \\ 1,2,3 \quad 7 \quad (a + c) + (b' + d') = (b + d) + (a' + c') = E(4, 5, 6) \end{array}$$

Schluss:

1	1	$a, b, a', b', c, d, c', d' \in \mathbb{N}$	A
2	2	$[a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}$	A
3	3	$[c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}}$	A
1,2,3	4	$a + b' = b + a' \wedge c + d' = d + c'$	Th. 17.2.4.5(i)(1,2,3)
1,2,3	5	$(a + c) + (b' + d') = (b + d) + (a' + c')$	Th. 17.2.4.5(ii)(1, $\wedge E1(4), \wedge E2(4)$)
1,2,3	6	$[a + c, b + d]_{\mathbb{Z}} = [a' + c', b' + d']_{\mathbb{Z}}$	Th. 17.2.2.2(1,5)

□

Theorem 17.2.4.6 (Quotientenabstieg der Addition).

Mengen: $z, w, u, v, a, b, c, d, a', b', c', d', p, q, r, s$

$$z, w \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}})$$

Beweis.

Ergebniskandidat fester Repräsentanten

Th. 17.2.4.6(i)

$$a, b, c, d \in \mathbb{N}, z = [a, b]_{\mathbb{Z}}, w = [c, d]_{\mathbb{Z}} \vdash \\ [a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [a + c, b + d]_{\mathbb{Z}} = [p + r, q + s]_{\mathbb{Z}})$$

1	1	$a, b, c, d \in \mathbb{N}$	
2	2	$z = [a, b]_{\mathbb{Z}}$	
3	3	$w = [c, d]_{\mathbb{Z}}$	
1	4	$a + c \in \mathbb{N}$	
1	5	$b + d \in \mathbb{N}$	
1	6	$[a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z}$	
	7	$[a + c, b + d]_{\mathbb{Z}} = [a + c, b + d]_{\mathbb{Z}}$	
1,2,3	8	$\exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [a + c, b + d]_{\mathbb{Z}} = [p + r, q + s]_{\mathbb{Z}})$	
1,2,3	9	$[a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [a + c, b + d]_{\mathbb{Z}} = [p + r, q + s]_{\mathbb{Z}})$	

Existenz durch getrennte Repräsentantenwahl

Th. 17.2.4.6(ii)

$$z, w \in \mathbb{Z} \vdash \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}})$$

- 1 1 $z \in \mathbb{Z}$
- 2 2 $w \in \mathbb{Z}$

$$\begin{array}{ll}
1 & 3 \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} \\
2 & 4 \exists c \in \mathbb{N} \exists d \in \mathbb{N} w = [c, d]_{\mathbb{Z}} \\
5 & 5 a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \\
6 & 6 c, d \in \mathbb{N} \wedge w = [c, d]_{\mathbb{Z}} \\
5 & 7 a, b \in \mathbb{N} \\
5 & 8 z = [a, b]_{\mathbb{Z}} \\
6 & 9 c, d \in \mathbb{N} \\
6 & 10 w = [c, d]_{\mathbb{Z}} \\
5,6 & 11 a, b, c, d \in \mathbb{N} \\
5,6 & 12 [a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [a + c, b + d]_{\mathbb{Z}} = [p + r, q + s]_{\mathbb{Z}}) \\
5,6 & 13 \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}}) \\
5 & 14 \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}}) \\
1,2 & 15 \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}})
\end{array}$$

Eindeutigkeit für feste Repräsentanten

Th. 17.2.4.6(iii)

$$\begin{aligned}
& a, b, c, d, a', b', c', d' \in \mathbb{N}, \\
& z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}}, \\
& z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}} \vdash u = v
\end{aligned}$$

1	1	$a, b, c, d, a', b', c', d' \in \mathbb{N}$	A
2	2	$z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}}$	A
3	3	$z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}}$	A
2	4	$z = [a, b]_{\mathbb{Z}}$	$\wedge E1(2)$
2	5	$w = [c, d]_{\mathbb{Z}}$	$\wedge E1(\wedge E2(2))$
2	6	$u = [a + c, b + d]_{\mathbb{Z}}$	$\wedge E2(\wedge E2(2))$
3	7	$z = [a', b']_{\mathbb{Z}}$	$\wedge E1(3)$
3	8	$w = [c', d']_{\mathbb{Z}}$	$\wedge E1(\wedge E2(3))$
3	9	$v = [a' + c', b' + d']_{\mathbb{Z}}$	$\wedge E2(\wedge E2(3))$
2,3	10	$[a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}$	$= E(4, 7)$
2,3	11	$[c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}}$	$= E(5, 8)$
1,2,3	12	$[a + c, b + d]_{\mathbb{Z}} = [a' + c', b' + d']_{\mathbb{Z}}$	Th. 17.2.4.5(1,10,11)
1,2,3	13	$u = v$	$= E(6, 9, 12)$

Eindeutigkeit beliebiger Ergebniszeugen

Th. 17.2.4.6(iv)

$$\begin{aligned}
& (u \in \mathbb{Z} \wedge \exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}})), \\
& (v \in \mathbb{Z} \wedge \exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}})) \vdash u = v
\end{aligned}$$

1	1	$u \in \mathbb{Z} \wedge \exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}})$	A
2	2	$v \in \mathbb{Z} \wedge \exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}})$	A

1	3	$\exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}})$	$\wedge E2(1)$
2	4	$\exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}})$	$\wedge E2(2)$
5	5	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}}$	A
6	6	$a', b', c', d' \in \mathbb{N} \wedge z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}}$	A
5	7	$a, b, c, d \in \mathbb{N}$	$\wedge E1(5)$
5	8	$z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}}$	$\wedge E2(5)$
6	9	$a', b', c', d' \in \mathbb{N}$	$\wedge E1(6)$
6	10	$z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}}$	$\wedge E2(6)$
5,6	11	$a, b, c, d, a', b', c', d' \in \mathbb{N}$	$\wedge I(7, 9)$
5,6	12	$u = v$	Th. 17.2.4.6(iii)(11,8,10)
5	13	$u = v$	$\exists E(4, 6, 12)$
1,2	14	$u = v$	$\exists E(3, 5, 13)$

Abschluss des Abstiegs

Th. 17.2.4.6(v)

$$z, w \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}})$$

1	1	$z, w \in \mathbb{Z}$	A
1	2	$\exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}})$	Th. 17.2.4.6(ii)(1)
3	3	$u \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}})$	A
4	4	$v \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge v = [p + r, q + s]_{\mathbb{Z}})$	A
3,4	5	$u = v$	Th. 17.2.4.6(iv)(3,4)
1	6	$\exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}})$	$\exists! I(2, 3, 4, 5)$

□

Der eindeutige Existenzsatz ist der Quotientenabstieg der Addition; die folgende Klassenregel bezeichnet dieses eindeutig bestimmte Ergebnis.

Definition 17.2.4.2 (Addition ganzer Zahlen).

Mengen: a, b, c, d
Neue Symbole: $+$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}} := [a + c, b + d]_{\mathbb{Z}}$$

Theorem 17.2.4.7 (Abgeschlossenheit der Addition ganzer Zahlen).

Mengen: z, w, a, b, c, d

$$z \in \mathbb{Z}, w \in \mathbb{Z} \vdash z + w \in \mathbb{Z}$$

Beweis.

Abgeschlossenheit für feste Repräsentanten

Th. 17.2.4.7(i)

$$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \\ \vdash z + w \in \mathbb{Z}$$

$$\begin{array}{ll} 1 & 1 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A \\ 1 & 2 \ z + w = [a + c, b + d]_{\mathbb{Z}} \quad \text{Def. 17.2.4.2(1)} \\ 1 & 3 \ a + c \in \mathbb{N} \quad \text{Th. 10.4.4.1(1)} \\ 1 & 4 \ b + d \in \mathbb{N} \quad \text{Th. 10.4.4.1(1)} \\ 1 & 5 \ [a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z} \quad \text{Th. 17.2.2.1(3,4)} \\ 1 & 6 \ z + w \in \mathbb{Z} \quad = E(2, 5) \end{array}$$

Elimination der Repräsentanten

Th. 17.2.4.7(ii)

$$z \in \mathbb{Z}, w \in \mathbb{Z} \vdash z + w \in \mathbb{Z}$$

$$\begin{array}{ll} 1 & 1 \ z \in \mathbb{Z} \quad A \\ 2 & 2 \ w \in \mathbb{Z} \quad A \\ 1 & 3 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}} \quad \text{Th. 17.2.2.3(1)} \\ 2 & 4 \ \exists c, d \in \mathbb{N} \ w = [c, d]_{\mathbb{Z}} \quad \text{Th. 17.2.2.3(2)} \\ 5 & 5 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A \\ 5 & 6 \ z + w \in \mathbb{Z} \quad \text{Th. 17.2.4.7(i)(5)} \\ 1,2 & 7 \ z + w \in \mathbb{Z} \quad \exists E(3, 4, 5, 6) \end{array}$$

□

Theorem 17.2.4.8 (Assoziativität der Addition ganzer Zahlen).

Mengen: $z, w, u, a, b, c, d, e, f$

$$z, w, u \in \mathbb{Z} \vdash (z + w) + u = z + (w + u)$$

Beweis.

Rechnung mit Repräsentanten

Th. 17.2.4.8(i)

$$a, b, c, d, e, f \in \mathbb{N} \vdash \\ ([a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}}) + [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} + ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}})$$

$$\begin{array}{ll} 1 & 1 \ a, b, c, d, e, f \in \mathbb{N} \quad A \\ 1 & 2 \ ([a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}}) + [e, f]_{\mathbb{Z}} = [(a + c) + e, (b + d) + f]_{\mathbb{Z}} \text{Def. 17.2.4.2(1)} \end{array}$$

$$\begin{array}{ll}
1 \ 3 \ [a, b]_{\mathbb{Z}} + ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a + (c + e), b + (d + f)]_{\mathbb{Z}} & \text{Def. 17.2.4.2(1)} \\
1 \ 4 \ (a + c) + e = a + (c + e) & \text{Th. 10.4.4.7(1)} \\
1 \ 5 \ (b + d) + f = b + (d + f) & \text{Th. 10.4.4.7(1)} \\
1 \ 6 \ [(a + c) + e, (b + d) + f]_{\mathbb{Z}} = [a + (c + e), b + (d + f)]_{\mathbb{Z}} = E(4, 5) \\
1 \ 7 \ ([a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}}) + [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} + ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = E(2, 3, 6)
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 \ 1 \ z, w, u \in \mathbb{Z} & A \\
1 \ 2 \ z \in \mathbb{Z} & \wedge E1(1) \\
1 \ 3 \ w \in \mathbb{Z} & \wedge E1(\wedge E2(1)) \\
1 \ 4 \ u \in \mathbb{Z} & \wedge E2(\wedge E2(1)) \\
1 \ 5 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}} & \text{Th. 17.2.2.3(2)} \\
1 \ 6 \ \exists c, d \in \mathbb{N} \ w = [c, d]_{\mathbb{Z}} & \text{Th. 17.2.2.3(3)} \\
1 \ 7 \ \exists e, f \in \mathbb{N} \ u = [e, f]_{\mathbb{Z}} & \text{Th. 17.2.2.3(4)} \\
8 \ 8 \ a, b, c, d, e, f \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [e, f]_{\mathbb{Z}} & A \\
8 \ 9 \ ([a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}}) + [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} + ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) & \text{Th. 17.2.4.8(i)(8)} \\
8 \ 10 \ (z + w) + u = z + (w + u) & = E(8, 9) \\
1 \ 11 \ (z + w) + u = z + (w + u) & \exists E(5, 6, 7, 8, 10)
\end{array}$$

□

Theorem 17.2.4.9 (Kommutativität der Addition ganzer Zahlen).

Mengen: z, w, a, b, c, d

$$z, w \in \mathbb{Z} \vdash z + w = w + z$$

Beweis.

Vertauschung fester Repräsentanten

Th. 17.2.4.9(i)

$$\begin{array}{l}
a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \\
\vdash z + w = w + z
\end{array}$$

$$\begin{array}{ll}
1 \ 1 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} & A \\
1 \ 2 \ z + w = [a + c, b + d]_{\mathbb{Z}} & \text{Def. 17.2.4.2(1)} \\
1 \ 3 \ w + z = [c + a, d + b]_{\mathbb{Z}} & \text{Def. 17.2.4.2(1)} \\
1 \ 4 \ a + c = c + a & \text{Th. 10.4.4.8(1)} \\
1 \ 5 \ b + d = d + b & \text{Th. 10.4.4.8(1)} \\
1 \ 6 \ [a + c, b + d]_{\mathbb{Z}} = [c + a, d + b]_{\mathbb{Z}} & = E(4, 5) \\
1 \ 7 \ z + w = w + z & = E(2, 3, 6)
\end{array}$$

Getrennte Repräsentantenwahl

Th. 17.2.4.9(ii)

$$z, w \in \mathbb{Z} \vdash z + w = w + z$$

1	$z, w \in \mathbb{Z}$	A
1	$z \in \mathbb{Z}$	$\wedge E1(1)$
1	$w \in \mathbb{Z}$	$\wedge E2(1)$
1	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(2)
1	$\exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 17.2.2.3(3)
6	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}}$	A
6	$z + w = w + z$	Th. 17.2.4.9(i)(6)
1	$z + w = w + z$	$\exists E(4, 5, 6, 7)$

□

Theorem 17.2.4.10 (Nullgesetze der Addition ganzer Zahlen).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash z + 0_{\mathbb{Z}} = z \wedge 0_{\mathbb{Z}} + z = z$$

Beweis.

Rechtsneutrales Element für einen Repräsentanten Th. 17.2.4.10(i)

$$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \\ \vdash z + 0_{\mathbb{Z}} = z$$

1	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
2	$0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}$	Th. 17.2.3.4
1	$z + 0_{\mathbb{Z}} = [a + 0, b + 0]_{\mathbb{Z}}$	Def. 17.2.4.2(1,2)
1	$a + 0 = a$	Th. 10.4.4.2(1)
1	$b + 0 = b$	Th. 10.4.4.2(1)
1	$z + 0_{\mathbb{Z}} = z$	$= E(1, 3, 4, 5)$

Elimination des Repräsentanten

Th. 17.2.4.10(ii)

$$z \in \mathbb{Z} \vdash z + 0_{\mathbb{Z}} = z \wedge 0_{\mathbb{Z}} + z = z$$

1	$z \in \mathbb{Z}$	A
1	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(1)
3	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3	$z + 0_{\mathbb{Z}} = z$	Th. 17.2.4.10(i)(3)

$$\begin{array}{ll}
3 \ 5 \ 0_{\mathbb{Z}} + z = z & \text{Th. 17.2.4.9(Th. 17.2.3.5,1, Th. 2.9.1.1(4))} \\
3 \ 6 \ z + 0_{\mathbb{Z}} = z \wedge 0_{\mathbb{Z}} + z = z \wedge I(4, 5) \\
1 \ 7 \ z + 0_{\mathbb{Z}} = z \wedge 0_{\mathbb{Z}} + z = z \exists E(2, 3, 6)
\end{array}$$

□

Theorem 17.2.4.11 (Additive Inversenregeln).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash z + (-z) = 0_{\mathbb{Z}} \wedge (-z) + z = 0_{\mathbb{Z}}$$

Beweis.

Inverse Rechnung auf einer Klasse

Th. 17.2.4.11(i)

$$a, b \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} + (-[a, b]_{\mathbb{Z}}) = 0_{\mathbb{Z}}$$

$$\begin{array}{ll}
1 \ 1 \ a, b \in \mathbb{N} & A \\
1 \ 2 \ -[a, b]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}} & \text{Def. 17.2.4.1(1)} \\
1 \ 3 \ [a, b]_{\mathbb{Z}} + (-[a, b]_{\mathbb{Z}}) = [a + b, b + a]_{\mathbb{Z}} & \text{Def. 17.2.4.2(1,2)} \\
1 \ 4 \ a + b = b + a & \text{Th. 10.4.4.8(1)} \\
1 \ 5 \ [a + b, b + a]_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}} & \text{Th. 17.2.2.2(Th. 10.4.4.1(1), Th. 10.4.4.1(1), Ax. 10.3.1, Ax. 10.3.1, = I)} \\
6 \ 0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}} & \text{Th. 17.2.3.4} \\
1 \ 7 \ [a, b]_{\mathbb{Z}} + (-[a, b]_{\mathbb{Z}}) = 0_{\mathbb{Z}} & = E(3, 5, 6)
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 \ 1 \ z \in \mathbb{Z} & A \\
1 \ 2 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}} & \text{Th. 17.2.2.3(1)} \\
3 \ 3 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} & A \\
3 \ 4 \ z + (-z) = 0_{\mathbb{Z}} & \text{Th. 17.2.4.11(i)(3)} \\
1 \ 5 \ -z \in \mathbb{Z} & \text{Th. 17.2.4.3(1)} \\
1,3 \ 6 \ (-z) + z = z + (-z) & \text{Th. 17.2.4.9(5,1)} \\
1,3 \ 7 \ (-z) + z = 0_{\mathbb{Z}} & = E(6, 4) \\
1,3 \ 8 \ z + (-z) = 0_{\mathbb{Z}} \wedge (-z) + z = 0_{\mathbb{Z}} \wedge I(4, 7) \\
1 \ 9 \ z + (-z) = 0_{\mathbb{Z}} \wedge (-z) + z = 0_{\mathbb{Z}} \exists E(2, 3, 8)
\end{array}$$

□

Theorem 17.2.4.12 (Negation der Ganzzahlnull).

Mengen: $0_{\mathbb{Z}}$

$$-0_{\mathbb{Z}} = 0_{\mathbb{Z}}$$

1	$0_{\mathbb{Z}} \in \mathbb{Z}$	Th. 17.2.3.5
2	$-0_{\mathbb{Z}} \in \mathbb{Z}$	Th. 17.2.4.3(1)
3	$(-0_{\mathbb{Z}}) + 0_{\mathbb{Z}} = -0_{\mathbb{Z}} \wedge E1(\text{Th. 17.2.4.10(2)})$	
4	$(-0_{\mathbb{Z}}) + 0_{\mathbb{Z}} = 0_{\mathbb{Z}} \wedge E2(\text{Th. 17.2.4.11(1)})$	
5	$-0_{\mathbb{Z}} = (-0_{\mathbb{Z}}) + 0_{\mathbb{Z}}$	Th. 2.9.1.1(3)
6	$-0_{\mathbb{Z}} = 0_{\mathbb{Z}}$	$= E(5, 4)$

2.5 Multiplikation

Theorem 17.2.5.1 (Repräsentantenunabhängigkeit der Multiplikation).

Mengen: $a, b, a', b', c, d, c', d'$

$$a, b, a', b', c, d, c', d' \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}, [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\ [a \cdot c + b \cdot d, a \cdot d + b \cdot c]_{\mathbb{Z}} = [a' \cdot c' + b' \cdot d', a' \cdot d' + b' \cdot c']_{\mathbb{Z}}$$

Beweis.

Wechsel des ersten Repräsentanten

Th. 17.2.5.1(i)

$$a, b, a', b', c, d \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}} \vdash \\ [ac + bd, ad + bc]_{\mathbb{Z}} = [a'c + b'd, a'd + b'c]_{\mathbb{Z}}$$

1	$a, b, a', b', c, d \in \mathbb{N}$	A
2	$[a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}$	A
1,2	$a + b' = b + a'$	Th. 17.2.2.2(1,2)
1,2	$(a + b')c = (b + a')c$	$= E(3)$
1	$(a + b')c = ac + b'c$	Th. 10.4.7.12(1)
1	$(b + a')c = bc + a'c$	Th. 10.4.7.12(1)
1,2	$ac + b'c = bc + a'c$	$= E(4, 5, 6)$
1,2	$(a + b')d = (b + a')d$	$= E(3)$
1	$(a + b')d = ad + b'd$	Th. 10.4.7.12(1)
1	$(b + a')d = bd + a'd$	Th. 10.4.7.12(1)
1,2	$ad + b'd = bd + a'd$	$= E(8, 9, 10)$
1	$a'd + b'c = b'c + a'd$	Th. 10.4.4.8(1)
1,2	$(ac + bd) + (a'd + b'c) = (ac + bd) + (b'c + a'd)$	$= E(12)$
1	$(ac + bd) + (b'c + a'd) = (ac + b'c) + (bd + a'd)$	Th. 10.4.4.10(1)
1,2	$bd + a'd = ad + b'd$	Th. 2.9.1.1(11)
1,2	$(ac + b'c) + (bd + a'd) = (bc + a'c) + (ad + b'd)$	$= E(7, 15)$
1	$(bc + a'c) + (ad + b'd) = (bc + ad) + (a'c + b'd)$	Th. 10.4.4.10(1)

$$\begin{array}{ll}
1 & 18 \ bc + ad = ad + bc \quad \text{Th. 10.4.4.8(1)} \\
1 & 19 \ (bc + ad) + (a'c + b'd) = (ad + bc) + (a'c + b'd) = E(18) \\
1,2 & 20 \ (ac + bd) + (a'd + b'c) = (ad + bc) + (a'c + b'd) = E(13, 14, 16, 17, 19) \\
1,2 & 21 \ [ac + bd, ad + bc]_{\mathbb{Z}} = [a'c + b'd, a'd + b'c]_{\mathbb{Z}} \quad \text{Th. 17.2.2.2(1,20)}
\end{array}$$

Wechsel des zweiten Repräsentanten

Th. 17.2.5.1(ii)

$$\begin{array}{l}
a, b, c, d, c', d' \in \mathbb{N}, \ [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\
[ac + bd, ad + bc]_{\mathbb{Z}} = [ac' + bd', ad' + bc']_{\mathbb{Z}}
\end{array}$$

$$\begin{array}{ll}
1 & 1 \ a, b, c, d, c', d' \in \mathbb{N} \quad A \\
2 & 2 \ [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \quad A \\
1,2 & 3 \ c + d' = d + c' \quad \text{Th. 17.2.2.2(1,2)} \\
1,2 & 4 \ a(c + d') = a(d + c') \quad = E(3) \\
1 & 5 \ a(c + d') = ac + ad' \quad \text{Th. 10.4.7.11(1)} \\
1 & 6 \ a(d + c') = ad + ac' \quad \text{Th. 10.4.7.11(1)} \\
1,2 & 7 \ ac + ad' = ad + ac' \quad = E(4, 5, 6) \\
1,2 & 8 \ b(c + d') = b(d + c') \quad = E(3) \\
1 & 9 \ b(c + d') = bc + bd' \quad \text{Th. 10.4.7.11(1)} \\
1 & 10 \ b(d + c') = bd + bc' \quad \text{Th. 10.4.7.11(1)} \\
1,2 & 11 \ bc + bd' = bd + bc' \quad = E(8, 9, 10) \\
1 & 12 \ (ac + bd) + (ad' + bc') = (ac + ad') + (bd + bc') \text{Th. 10.4.4.10(1)} \\
1,2 & 13 \ bd + bc' = bc + bd' \quad \text{Th. 2.9.1.1(11)} \\
1,2 & 14 \ (ac + ad') + (bd + bc') = (ad + ac') + (bc + bd') = E(7, 13) \\
1 & 15 \ (ad + ac') + (bc + bd') = (ad + bc) + (ac' + bd') \text{Th. 10.4.4.10(1)} \\
1,2 & 16 \ (ac + bd) + (ad' + bc') = (ad + bc) + (ac' + bd') = E(12, 14, 15) \\
1,2 & 17 \ [ac + bd, ad + bc]_{\mathbb{Z}} = [ac' + bd', ad' + bc']_{\mathbb{Z}} \quad \text{Th. 17.2.2.2(1,16)}
\end{array}$$

Zusammenführung:

$$\begin{array}{ll}
1 & 1 \ a, b, a', b', c, d, c', d' \in \mathbb{N} \quad A \\
2 & 2 \ [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}} \quad A \\
3 & 3 \ [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \quad A \\
1,2 & 4 \ [ac + bd, ad + bc]_{\mathbb{Z}} = [a'c + b'd, a'd + b'c]_{\mathbb{Z}} \quad \text{Th. 17.2.5.1(i)(1,2)} \\
1,3 & 5 \ [a'c + b'd, a'd + b'c]_{\mathbb{Z}} = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}} \text{Th. 17.2.5.1(ii)(1,3)} \\
1,2,3 & 6 \ [ac + bd, ad + bc]_{\mathbb{Z}} = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}} \quad \text{Th. 2.9.1.2(4,5)}
\end{array}$$

□

Theorem 17.2.5.2 (Quotientenabstieg der Multiplikation).

Mengen: $z, w, u, v, a, b, c, d, a', b', c', d', p, q, r, s$

$$z, w \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})$$

Beweis.

Abgeschlossenheit der Ergebniskoordinaten

Th. 17.2.5.2(i)

$$a, b, c, d \in \mathbb{N} \vdash ac + bd \in \mathbb{N} \wedge ad + bc \in \mathbb{N}$$

1	1	$a, b, c, d \in \mathbb{N}$	A
1	2	$ac \in \mathbb{N}$	Th. 10.4.7.3(1)
1	3	$bd \in \mathbb{N}$	Th. 10.4.7.3(1)
1	4	$ad \in \mathbb{N}$	Th. 10.4.7.3(1)
1	5	$bc \in \mathbb{N}$	Th. 10.4.7.3(1)
1	6	$ac + bd \in \mathbb{N}$	Th. 10.4.4.1(2,3)
1	7	$ad + bc \in \mathbb{N}$	Th. 10.4.4.1(4,5)
1	8	$ac + bd \in \mathbb{N} \wedge ad + bc \in \mathbb{N}$	$\wedge I(6, 7)$

Ergebniskandidat fester Repräsentanten

Th. 17.2.5.2(ii)

$$a, b, c, d \in \mathbb{N}, \quad z = [a, b]_{\mathbb{Z}}, \quad w = [c, d]_{\mathbb{Z}} \vdash \\ [ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [ac + bd, ad + bc]_{\mathbb{Z}} = [pr + qs, ps + qr]_{\mathbb{Z}})$$

1	1	$a, b, c, d \in \mathbb{N}$	
2	2	$z = [a, b]_{\mathbb{Z}}$	
3	3	$w = [c, d]_{\mathbb{Z}}$	
1	4	$ac + bd \in \mathbb{N} \wedge ad + bc \in \mathbb{N}$	
1	5	$[ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z}$	
	6	$[ac + bd, ad + bc]_{\mathbb{Z}} = [ac + bd, ad + bc]_{\mathbb{Z}}$	
1,2,3	7	$\exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [ac + bd, ad + bc]_{\mathbb{Z}} = [pr + qs, ps + qr]_{\mathbb{Z}})$	
1,2,3	8	$[ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [ac + bd, ad + bc]_{\mathbb{Z}} = [pr + qs, ps + qr]_{\mathbb{Z}})$	

Existenz durch getrennte Repräsentantenwahl

Th. 17.2.5.2(iii)

$$z, w \in \mathbb{Z} \vdash \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})$$

1	1	$z, w \in \mathbb{Z}$
1	2	$z \in \mathbb{Z}$
1	3	$w \in \mathbb{Z}$

$$\begin{array}{ll}
1 & 4 \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} \\
1 & 5 \exists c \in \mathbb{N} \exists d \in \mathbb{N} w = [c, d]_{\mathbb{Z}} \\
6 & 6 a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \\
7 & 7 c, d \in \mathbb{N} \wedge w = [c, d]_{\mathbb{Z}} \\
6 & 8 a, b \in \mathbb{N} \\
6 & 9 z = [a, b]_{\mathbb{Z}} \\
7 & 10 c, d \in \mathbb{N} \\
7 & 11 w = [c, d]_{\mathbb{Z}} \\
6,7 & 12 a, b, c, d \in \mathbb{N} \\
6,7 & 13 [ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [ac + bd, ad + bc]_{\mathbb{Z}} = [pr + qs, ps + qr]_{\mathbb{Z}}) \\
6,7 & 14 \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}}) \\
6 & 15 \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}}) \\
1 & 16 \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})
\end{array}$$

Eindeutigkeit für feste Repräsentanten

Th. 17.2.5.2(iv)

$$\begin{aligned}
& a, b, c, d, a', b', c', d' \in \mathbb{N}, \\
& z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}}, \\
& z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}} \vdash u = v
\end{aligned}$$

1	1	$a, b, c, d, a', b', c', d' \in \mathbb{N}$	A
2	2	$z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}}$	A
3	3	$z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	A
2	4	$z = [a, b]_{\mathbb{Z}}$	$\wedge E1(2)$
2	5	$w = [c, d]_{\mathbb{Z}}$	$\wedge E1(\wedge E2(2))$
2	6	$u = [ac + bd, ad + bc]_{\mathbb{Z}}$	$\wedge E2(\wedge E2(2))$
3	7	$z = [a', b']_{\mathbb{Z}}$	$\wedge E1(3)$
3	8	$w = [c', d']_{\mathbb{Z}}$	$\wedge E1(\wedge E2(3))$
3	9	$v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	$\wedge E2(\wedge E2(3))$
2,3	10	$[a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}$	$= E(4, 7)$
2,3	11	$[c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}}$	$= E(5, 8)$
1,2,3	12	$[ac + bd, ad + bc]_{\mathbb{Z}} = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	Th. 17.2.5.1(1,10,11)
1,2,3	13	$u = v$	$= E(6, 9, 12)$

Eindeutigkeit beliebiger Ergebniszeugen

Th. 17.2.5.2(v)

$$\begin{aligned}
& (u \in \mathbb{Z} \wedge \exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}})), \\
& (v \in \mathbb{Z} \wedge \exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}})) \vdash u = v
\end{aligned}$$

1	1	$u \in \mathbb{Z} \wedge \exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}})$	A
2	2	$v \in \mathbb{Z} \wedge \exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}})$	A

1	3	$\exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}})$	$\wedge E2(1)$
2	4	$\exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}})$	$\wedge E2(2)$
5	5	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}}$	A
6	6	$a', b', c', d' \in \mathbb{N} \wedge z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	A
5	7	$a, b, c, d \in \mathbb{N}$	$\wedge E1(5)$
5	8	$z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}}$	$\wedge E2(5)$
6	9	$a', b', c', d' \in \mathbb{N}$	$\wedge E1(6)$
6	10	$z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	$\wedge E2(6)$
5,6	11	$a, b, c, d, a', b', c', d' \in \mathbb{N}$	$\wedge I(7, 9)$
5,6	12	$u = v$	Th. 17.2.5.2(iv)(11, 8, 1)
5	13	$u = v$	$\exists E(4, 6, 12)$
1,2	14	$u = v$	$\exists E(3, 5, 13)$

Abschluss des Abstiegs

Th. 17.2.5.2(vi)

$$z, w \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})$$

1	1	$z, w \in \mathbb{Z}$	A
1	2	$\exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})$	Th. 17.2.5.2(iii)(1)
3	3	$u \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})$	A
4	4	$v \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge v = [pr + qs, ps + qr]_{\mathbb{Z}})$	A
3,4	5	$u = v$	Th. 17.2.5.2(v)(3, 4)
1	6	$\exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})$	$\exists! I(2, 3, 4, 5)$

□

Der Quotientenabstieg liefert damit auch für die Multiplikation unabhängig von beiden gewählten Differenzenpaaren genau ein Ergebnis in $\mathbb{Z}$.

Definition 17.2.5.1 (Multiplikation ganzer Zahlen).

Mengen: a, b, c, d
Neue Symbole: $\cdot$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} := [a \cdot c + b \cdot d, a \cdot d + b \cdot c]_{\mathbb{Z}}$$

Theorem 17.2.5.3 (Abgeschlossenheit der Multiplikation ganzer Zahlen).

Mengen: z, w, a, b, c, d

$$z \in \mathbb{Z}, w \in \mathbb{Z} \vdash z \cdot w \in \mathbb{Z}$$

Beweis.

Abgeschlossenheit für feste Repräsentanten

Th. 17.2.5.3(i)

$$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \\ \vdash z \cdot w \in \mathbb{Z}$$

1	$1 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A$	
1	$2 \ z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}}$	Def. 17.2.5.1(1)
1	$3 \ ac + bd \in \mathbb{N}$	Th. 10.4.4.1(Th. 10.4.7.3(1), Th. 10.4.7.3(1))
1	$4 \ ad + bc \in \mathbb{N}$	Th. 10.4.4.1(Th. 10.4.7.3(1), Th. 10.4.7.3(1))
1	$5 \ [ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z}$	Th. 17.2.2.1(3,4)
1	$6 \ z \cdot w \in \mathbb{Z}$	$= E(2, 5)$

Elimination der Repräsentanten

Th. 17.2.5.3(ii)

$$z \in \mathbb{Z}, w \in \mathbb{Z} \vdash z \cdot w \in \mathbb{Z}$$

1	$1 \ z \in \mathbb{Z}$	A
2	$2 \ w \in \mathbb{Z}$	A
1	$3 \ \exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(1)
2	$4 \ \exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 17.2.2.3(2)
5	$5 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A$	
5	$6 \ z \cdot w \in \mathbb{Z}$	Th. 17.2.5.3(i)(5)
1,2	$7 \ z \cdot w \in \mathbb{Z}$	$\exists E(3, 4, 5, 6)$

□

Theorem 17.2.5.4 (Kommutativität der Multiplikation ganzer Zahlen).

Mengen: z, w, a, b, c, d

$$z, w \in \mathbb{Z} \vdash z \cdot w = w \cdot z$$

Beweis.

Produkte gewählter Repräsentanten

Th. 17.2.5.4(i)

$$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \\ \vdash z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}} \wedge w \cdot z = [ca + db, cb + da]_{\mathbb{Z}}$$

1	1	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}}$	A
1	2	$z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}}$	Def. 17.2.5.1(1)
1	3	$w \cdot z = [ca + db, cb + da]_{\mathbb{Z}}$	Def. 17.2.5.1(1)
1	4	$z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}} \wedge w \cdot z = [ca + db, cb + da]_{\mathbb{Z}} \wedge I(2, 3)$	

Vertauschung der Klassenkoordinaten

Th. 17.2.5.4(ii)

$$a, b, c, d \in \mathbb{N} \vdash [ac + bd, ad + bc]_{\mathbb{Z}} = [ca + db, cb + da]_{\mathbb{Z}}$$

1	1	$a, b, c, d \in \mathbb{N}$	A
1	2	$ac = ca$	Th. 10.4.7.9(1)
1	3	$bd = db$	Th. 10.4.7.9(1)
1	4	$ad = da$	Th. 10.4.7.9(1)
1	5	$bc = cb$	Th. 10.4.7.9(1)
1	6	$ac + bd = ca + db$	$= E(2, 3)$
1	7	$ad + bc = cb + da$	Th. 10.4.4.8(= $E(4, 5)$)
1	8	$[ac + bd, ad + bc]_{\mathbb{Z}} = [ca + db, cb + da]_{\mathbb{Z}} = E(6, 7)$	

Elimination der Repräsentanten

Th. 17.2.5.4(iii)

$$z, w \in \mathbb{Z} \vdash z \cdot w = w \cdot z$$

1	1	$z, w \in \mathbb{Z}$	A
1	2	$z \in \mathbb{Z}$	$\wedge E1(1)$
1	3	$w \in \mathbb{Z}$	$\wedge E2(1)$
1	4	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(2)
1	5	$\exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 17.2.2.3(3)
6	6	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}}$	A
6	7	$z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}} \wedge w \cdot z = [ca + db, cb + da]_{\mathbb{Z}}$	Th. 17.2.5.4(i)(6)
6	8	$z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}}$	$\wedge E1(7)$
6	9	$w \cdot z = [ca + db, cb + da]_{\mathbb{Z}}$	$\wedge E2(7)$
6	10	$[ac + bd, ad + bc]_{\mathbb{Z}} = [ca + db, cb + da]_{\mathbb{Z}}$	Th. 17.2.5.4(ii)(6)
6	11	$z \cdot w = w \cdot z$	$= E(8, 9, 10)$
1	12	$z \cdot w = w \cdot z$	$\exists E(4, 5, 6, 11)$

□

Theorem 17.2.5.5 (Einheitsgesetze der Multiplikation ganzer Zahlen).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash z \cdot 1_{\mathbb{Z}} = z \wedge 1_{\mathbb{Z}} \cdot z = z$$

Beweis.

Rechte Einheit am Repräsentanten

Th. 17.2.5.5(i)

$$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \vdash z \cdot 1_{\mathbb{Z}} = z$$

1	$1 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
2	$1_{\mathbb{Z}} = \iota_{\mathbb{Z}}(1)$	Def. 17.2.3.3
3	$\iota_{\mathbb{Z}}(1) = [1, 0]_{\mathbb{Z}}$	Th. 17.2.3.2(Th. 10.4.4.11)
1	$4 \ z \cdot 1_{\mathbb{Z}} = [a \cdot 1 + b \cdot 0, a \cdot 0 + b \cdot 1]_{\mathbb{Z}}$	Def. 17.2.5.1(1,2,3)
1	$5 \ a \cdot 1 = a$	Th. 10.4.7.7(1)
1	$6 \ b \cdot 1 = b$	Th. 10.4.7.7(1)
1	$7 \ a \cdot 0 = 0$	Th. 10.4.7.4(1)
1	$8 \ b \cdot 0 = 0$	Th. 10.4.7.4(1)
1	$9 \ a \cdot 1 + b \cdot 0 = a$	$= E(5, 8, Th. 10.4.4.2(1))$
1	$10 \ a \cdot 0 + b \cdot 1 = b$	$= E(7, 6, Th. 10.4.4.4(1))$
1	$11 \ z \cdot 1_{\mathbb{Z}} = z$	$= E(1, 4, 9, 10)$

Beidseitige Einheit

Th. 17.2.5.5(ii)

$$z \in \mathbb{Z} \vdash z \cdot 1_{\mathbb{Z}} = z \wedge 1_{\mathbb{Z}} \cdot z = z$$

1	$1 \ z \in \mathbb{Z}$	A
1	$2 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(1)
3	$3 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3	$4 \ z \cdot 1_{\mathbb{Z}} = z$	Th. 17.2.5.5(i)(3)
3	$5 \ 1_{\mathbb{Z}} \cdot z = z$	Th. 17.2.5.4(Th. 17.2.3.6, 1, Th. 2.9.1.1(4))
3	$6 \ z \cdot 1_{\mathbb{Z}} = z \wedge 1_{\mathbb{Z}} \cdot z = z$	$\wedge I(4, 5)$
1	$7 \ z \cdot 1_{\mathbb{Z}} = z \wedge 1_{\mathbb{Z}} \cdot z = z$	$\exists E(2, 3, 6)$

□

Theorem 17.2.5.6 (Assoziativität der Multiplikation ganzer Zahlen).

Mengen: $z, w, u, a, b, c, d, e, f$

$$z, w, u \in \mathbb{Z} \vdash (z \cdot w) \cdot u = z \cdot (w \cdot u)$$

Beweis.

Positive Koordinate: linke Seite ausmultiplizieren

Th. 17.2.5.6(i)

$$\begin{aligned} a, b, c, d, e, f \in \mathbb{N} \vdash \\ (ac + bd)e + (ad + bc)f = ((ac)e + (bd)e) + ((ad)f + (bc)f) \end{aligned}$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$(ac + bd)e = (ac)e + (bd)e$	Th. 10.4.7.12(1)
1	3	$(ad + bc)f = (ad)f + (bc)f$	Th. 10.4.7.12(1)
1	4	$(ac + bd)e + (ad + bc)f = ((ac)e + (bd)e) + ((ad)f + (bc)f) = E(2, 3)$	

Positive Koordinate: rechte Seite ausmultiplizieren Th. 17.2.5.6(ii)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$a(ce + df) + b(cf + de) = (a(ce) + a(df)) + (b(cf) + b(de))$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$a(ce + df) = a(ce) + a(df)$	Th. 10.4.7.11(1)
1	3	$b(cf + de) = b(cf) + b(de)$	Th. 10.4.7.11(1)
1	4	$a(ce + df) + b(cf + de) = (a(ce) + a(df)) + (b(cf) + b(de)) = E(2, 3)$	

Positive Koordinate: ausmultiplizierte Terme ordnen Th. 17.2.5.6(iii)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$((ac)e + (bd)e) + ((ad)f + (bc)f)$$

$$= (a(ce) + a(df)) + (b(cf) + b(de))$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$(ac)e = a(ce)$	Th. 10.4.7.13(1)
1	3	$(bd)e = b(de)$	Th. 10.4.7.13(1)
1	4	$(ad)f = a(df)$	Th. 10.4.7.13(1)
1	5	$(bc)f = b(cf)$	Th. 10.4.7.13(1)
1	6	$((ac)e + (bd)e) + ((ad)f + (bc)f) = (a(ce) + b(de)) + (a(df) + b(cf)) = E(2, 3, 4, 5)$	
1	7	$(a(ce) + b(de)) + (a(df) + b(cf)) = (a(ce) + a(df)) + (b(de) + b(cf))$	Th. 10.4.4.10(1)
1	8	$b(de) + b(cf) = b(cf) + b(de)$	Th. 10.4.4.8(1)
1	9	$(a(ce) + a(df)) + (b(de) + b(cf)) = (a(ce) + a(df)) + (b(cf) + b(de)) = E(8)$	
1	10	$((ac)e + (bd)e) + ((ad)f + (bc)f) = (a(ce) + a(df)) + (b(cf) + b(de)) = E(6, 7, 9)$	

Gleichheit der positiven Koordinaten Th. 17.2.5.6(iv)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$(ac + bd)e + (ad + bc)f = a(ce + df) + b(cf + de)$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$(ac + bd)e + (ad + bc)f = ((ac)e + (bd)e) + ((ad)f + (bc)f)$	Th. 17.2.5.6(i)(1)
1	3	$((ac)e + (bd)e) + ((ad)f + (bc)f) = (a(ce) + a(df)) + (b(cf) + b(de))$	Th. 17.2.5.6(iii)(1)
1	4	$a(ce + df) + b(cf + de) = (a(ce) + a(df)) + (b(cf) + b(de))$	Th. 17.2.5.6(ii)(1)
1	5	$(a(ce) + a(df)) + (b(cf) + b(de)) = a(ce + df) + b(cf + de)$	Th. 2.9.1.1(4)

$$1 \quad 6 \quad (ac + bd)e + (ad + bc)f = a(cf + de) + b(cf + de) = E(2, 3, 5)$$

Negative Koordinate: linke Seite ausmultiplizieren Th. 17.2.5.6(v)

$$a, b, c, d, e, f \in \mathbb{N} \vdash \\ (ac + bd)f + (ad + bc)e = ((ac)f + (bd)f) + ((ad)e + (bc)e)$$

$$\begin{array}{ll} 1 \quad 1 \quad a, b, c, d, e, f \in \mathbb{N} & A \\ 1 \quad 2 \quad (ac + bd)f = (ac)f + (bd)f & \text{Th. 10.4.7.12(1)} \\ 1 \quad 3 \quad (ad + bc)e = (ad)e + (bc)e & \text{Th. 10.4.7.12(1)} \\ 1 \quad 4 \quad (ac + bd)f + (ad + bc)e = ((ac)f + (bd)f) + ((ad)e + (bc)e) = E(2, 3) \end{array}$$

Negative Koordinate: rechte Seite ausmultiplizieren Th. 17.2.5.6(vi)

$$a, b, c, d, e, f \in \mathbb{N} \vdash \\ a(cf + de) + b(ce + df) = (a(cf) + a(de)) + (b(ce) + b(df))$$

$$\begin{array}{ll} 1 \quad 1 \quad a, b, c, d, e, f \in \mathbb{N} & A \\ 1 \quad 2 \quad a(cf + de) = a(cf) + a(de) & \text{Th. 10.4.7.11(1)} \\ 1 \quad 3 \quad b(ce + df) = b(ce) + b(df) & \text{Th. 10.4.7.11(1)} \\ 1 \quad 4 \quad a(cf + de) + b(ce + df) = (a(cf) + a(de)) + (b(ce) + b(df)) = E(2, 3) \end{array}$$

Negative Koordinate: ausmultiplizierte Terme ordnen Th. 17.2.5.6(vii)

$$a, b, c, d, e, f \in \mathbb{N} \vdash \\ ((ac)f + (bd)f) + ((ad)e + (bc)e) \\ = (a(cf) + a(de)) + (b(ce) + b(df))$$

$$\begin{array}{ll} 1 \quad 1 \quad a, b, c, d, e, f \in \mathbb{N} & A \\ 1 \quad 2 \quad (ac)f = a(cf) & \text{Th. 10.4.7.13(1)} \\ 1 \quad 3 \quad (bd)f = b(df) & \text{Th. 10.4.7.13(1)} \\ 1 \quad 4 \quad (ad)e = a(de) & \text{Th. 10.4.7.13(1)} \\ 1 \quad 5 \quad (bc)e = b(ce) & \text{Th. 10.4.7.13(1)} \\ 1 \quad 6 \quad ((ac)f + (bd)f) + ((ad)e + (bc)e) = (a(cf) + b(df)) + (a(de) + b(ce)) = E(2, 3, 4, 5) \\ 1 \quad 7 \quad (a(cf) + b(df)) + (a(de) + b(ce)) = (a(cf) + a(de)) + (b(df) + b(ce)) \text{ Th. 10.4.4.10(1)} \\ 1 \quad 8 \quad b(df) + b(ce) = b(ce) + b(df) & \text{Th. 10.4.4.8(1)} \\ 1 \quad 9 \quad (a(cf) + a(de)) + (b(df) + b(ce)) = (a(cf) + a(de)) + (b(ce) + b(df)) = E(8) \\ 1 \quad 10 \quad ((ac)f + (bd)f) + ((ad)e + (bc)e) = (a(cf) + a(de)) + (b(ce) + b(df)) = E(6, 7, 9) \end{array}$$

Gleichheit der negativen Koordinaten Th. 17.2.5.6(viii)

$$a, b, c, d, e, f \in \mathbb{N} \vdash \\ (ac + bd)f + (ad + bc)e = a(cf + de) + b(ce + df)$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$(ac + bd)f + (ad + bc)e = ((ac)f + (bd)f) + ((ad)e + (bc)e)$	Th. 17.2.5.6(v)(1)
1	3	$((ac)f + (bd)f) + ((ad)e + (bc)e) = (a(cf) + a(de)) + (b(ce) + b(df))$	Th. 17.2.5.6(vii)(1)
1	4	$a(cf + de) + b(ce + df) = (a(cf) + a(de)) + (b(ce) + b(df))$	Th. 17.2.5.6(vi)(1)
1	5	$(a(cf) + a(de)) + (b(ce) + b(df)) = a(cf + de) + b(ce + df)$	Th. 2.9.1.1(4)
1	6	$(ac + bd)f + (ad + bc)e = a(cf + de) + b(ce + df)$	$= E(2, 3, 5)$

Rechnung mit Klassen

Th. 17.2.5.6(ix)

$$a, b, c, d, e, f \in \mathbb{N} \vdash \\ ([a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}}) \cdot [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}})$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$([a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}}) \cdot [e, f]_{\mathbb{Z}} = [(ac + bd)e + (ad + bc)f, (ac + bd)f + (ad + bc)e]_{\mathbb{Z}}$	Def. 17.2.5.1(1)
1	3	$[a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}}) = [a(ce + df) + b(cf + de), a(cf + de) + b(ce + df)]_{\mathbb{Z}}$	Def. 17.2.5.1(1)
1	4	$(ac + bd)e + (ad + bc)f = a(ce + df) + b(cf + de)$	Th. 17.2.5.6(iv)(1)
1	5	$(ac + bd)f + (ad + bc)e = a(cf + de) + b(ce + df)$	Th. 17.2.5.6(viii)(1)
1	6	$([a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}}) \cdot [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}})$	$= E(2, 3, 4, 5)$

Schluss:

1	1	$z, w, u \in \mathbb{Z}$	A
1	2	$z \in \mathbb{Z}$	$\wedge E1(1)$
1	3	$w \in \mathbb{Z}$	$\wedge E1(\wedge E2(1))$
1	4	$u \in \mathbb{Z}$	$\wedge E2(\wedge E2(1))$
1	5	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(2)
1	6	$\exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 17.2.2.3(3)
1	7	$\exists e, f \in \mathbb{N} u = [e, f]_{\mathbb{Z}}$	Th. 17.2.2.3(4)
8	8	$a, b, c, d, e, f \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [e, f]_{\mathbb{Z}}$	A
8	9	$([a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}}) \cdot [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}})$	Th. 17.2.5.6(ix)(8)
8	10	$(z \cdot w) \cdot u = z \cdot (w \cdot u)$	$= E(8, 9)$
1	11	$(z \cdot w) \cdot u = z \cdot (w \cdot u)$	$\exists E(5, 6, 7, 8, 10)$

□

Theorem 17.2.5.7 (Links distributivität ganzer Zahlen).

Mengen: $z, w, u, a, b, c, d, e, f$

$$z, w, u \in \mathbb{Z} \vdash z \cdot (w + u) = z \cdot w + z \cdot u$$

Beweis.

Positive Koordinate distributiv ausmultiplizieren Th. 17.2.5.7(i)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$a(c + e) + b(d + f) = (ac + bd) + (ae + bf)$$

1	1 $a, b, c, d, e, f \in \mathbb{N}$	A
1	2 $a(c + e) = ac + ae$	Th. 10.4.7.11(1)
1	3 $b(d + f) = bd + bf$	Th. 10.4.7.11(1)
1	4 $a(c + e) + b(d + f) = (ac + ae) + (bd + bf)$	$= E(2, 3)$
1	5 $(ac + ae) + (bd + bf) = (ac + bd) + (ae + bf)$	Th. 10.4.4.10(1)
1	6 $a(c + e) + b(d + f) = (ac + bd) + (ae + bf)$	$= E(4, 5)$

Negative Koordinate distributiv ausmultiplizieren Th. 17.2.5.7(ii)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$a(d + f) + b(c + e) = (ad + bc) + (af + be)$$

1	1 $a, b, c, d, e, f \in \mathbb{N}$	A
1	2 $a(d + f) = ad + af$	Th. 10.4.7.11(1)
1	3 $b(c + e) = bc + be$	Th. 10.4.7.11(1)
1	4 $a(d + f) + b(c + e) = (ad + af) + (bc + be)$	$= E(2, 3)$
1	5 $(ad + af) + (bc + be) = (ad + bc) + (af + be)$	Th. 10.4.4.10(1)
1	6 $a(d + f) + b(c + e) = (ad + bc) + (af + be)$	$= E(4, 5)$

Distributivrechnung auf Klassen Th. 17.2.5.7(iii)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$[a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} + [a, b]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}}$$

1	1 $a, b, c, d, e, f \in \mathbb{N}$	A
1	2 $[a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a(c + e) + b(d + f), a(d + f) + b(c + e)]_{\mathbb{Z}}$	Def. 17.2.4.2(Def. 17.2.5.1(1))
1	3 $[a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} + [a, b]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}} = [(ac + bd) + (ae + bf), (ad + bc) + (af + be)]_{\mathbb{Z}}$	Def. 17.2.5.1(Def. 17.2.4.2(1))
1	4 $a(c + e) + b(d + f) = (ac + bd) + (ae + bf)$	Th. 17.2.5.7(i)(1)
1	5 $a(d + f) + b(c + e) = (ad + bc) + (af + be)$	Th. 17.2.5.7(ii)(1)
1	6 $[a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} + [a, b]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}}$	$= E(2, 3, 4, 5)$

Schluss:

1	1 $z, w, u \in \mathbb{Z}$	A
1	2 $z \in \mathbb{Z}$	$\wedge E1(1)$
1	3 $w \in \mathbb{Z}$	$\wedge E1(\wedge E2(1))$

1	4	$u \in \mathbb{Z}$	$\wedge E2(\wedge E2(1))$
1	5	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(2)
1	6	$\exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 17.2.2.3(3)
1	7	$\exists e, f \in \mathbb{N} u = [e, f]_{\mathbb{Z}}$	Th. 17.2.2.3(4)
8	8	$a, b, c, d, e, f \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [e, f]_{\mathbb{Z}}$	A
8	9	$[a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} + [a, b]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}}$	Th. 17.2.5.7(iii)(8)
8	10	$z \cdot (w + u) = z \cdot w + z \cdot u$	$= E(8, 9)$
1	11	$z \cdot (w + u) = z \cdot w + z \cdot u$	$\exists E(5, 6, 7, 8, 10)$

□

Theorem 17.2.5.8 (Rechtsdistributivität ganzer Zahlen).

Mengen: z, w, u

$$z, w, u \in \mathbb{Z} \vdash (z + w) \cdot u = z \cdot u + w \cdot u$$

1	1	$z, w, u \in \mathbb{Z}$	A
1	2	$(z + w) \cdot u = u \cdot (z + w)$	Th. 17.2.5.4(Th. 17.2.4.7(1), 1)
1	3	$u \cdot (z + w) = u \cdot z + u \cdot w$	Th. 17.2.5.7(1)
1	4	$u \cdot z = z \cdot u$	Th. 17.2.5.4(1)
1	5	$u \cdot w = w \cdot u$	Th. 17.2.5.4(1)
1	6	$u \cdot z + u \cdot w = z \cdot u + w \cdot u$	$= E(4, 5)$
1	7	$(z + w) \cdot u = z \cdot u + w \cdot u$	$= E(2, 3, 6)$

2.6 Verträglichkeit der Einbettung mit der Arithmetik

Theorem 17.2.6.1 (Additivität der natürlichen Einbettung).

Mengen: m, n

Funktion: $\iota_{\mathbb{Z}}$

$$m, n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(m + n) = \iota_{\mathbb{Z}}(m) + \iota_{\mathbb{Z}}(n)$$

1	1	$m, n \in \mathbb{N}$	A
1	2	$m + n \in \mathbb{N}$	Th. 10.4.4.1(1)
1	3	$\iota_{\mathbb{Z}}(m + n) = [m + n, 0]_{\mathbb{Z}}$	Th. 17.2.3.2(2)
1	4	$\iota_{\mathbb{Z}}(m) = [m, 0]_{\mathbb{Z}}$	Th. 17.2.3.2(1)
1	5	$\iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}$	Th. 17.2.3.2(1)
1	6	$\iota_{\mathbb{Z}}(m) + \iota_{\mathbb{Z}}(n) = [m + n, 0 + 0]_{\mathbb{Z}}$	Def. 17.2.4.2(1, 4, 5)

$$\begin{array}{ll}
7 & 0 + 0 = 0 & Th. 10.4.4.2(Ax. 10.3.1) \\
1 & 8 \quad \iota_{\mathbb{Z}}(m) + \iota_{\mathbb{Z}}(n) = [m + n, 0]_{\mathbb{Z}} & = E(7, 6) \\
1 & 9 \quad \iota_{\mathbb{Z}}(m + n) = \iota_{\mathbb{Z}}(m) + \iota_{\mathbb{Z}}(n) & Th. 2.9.1.3(3, 8)
\end{array}$$

Theorem 17.2.6.2 (Multiplikativität der natürlichen Einbettung).

Mengen: m, n

Funktion: $\iota_{\mathbb{Z}}$

$$m, n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(m \cdot n) = \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n)$$

Beweis.

Produkt der eingebetteten Faktoren

Th. 17.2.6.2(i)

$$m, n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n) = [mn, 0]_{\mathbb{Z}}$$

$$\begin{array}{ll}
1 & 1 \quad m, n \in \mathbb{N} & A \\
1 & 2 \quad mn \in \mathbb{N} & Th. 10.4.7.3(1) \\
1 & 3 \quad \iota_{\mathbb{Z}}(m) = [m, 0]_{\mathbb{Z}} & Th. 17.2.3.2(1) \\
1 & 4 \quad \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}} & Th. 17.2.3.2(1) \\
1 & 5 \quad \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n) = [mn + 0 \cdot 0, m \cdot 0 + 0 \cdot n]_{\mathbb{Z}} & Def. 17.2.5.1(1, 3, 4) \\
& 6 \quad 0 \cdot 0 = 0 & Th. 10.4.7.6(Ax. 10.3.1) \\
1 & 7 \quad m \cdot 0 = 0 & Th. 10.4.7.4(1) \\
1 & 8 \quad 0 \cdot n = 0 & Th. 10.4.7.6(1) \\
1 & 9 \quad mn + 0 \cdot 0 = mn & = E(6, Th. 10.4.4.2(2)) \\
1 & 10 \quad m \cdot 0 + 0 \cdot n = 0 & = E(7, 8, Th. 10.4.4.2(Ax. 10.3.1)) \\
1 & 11 \quad \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n) = [mn, 0]_{\mathbb{Z}} & = E(5, 9, 10)
\end{array}$$

Vergleich mit dem Produktbild

Th. 17.2.6.2(ii)

$$m, n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(m \cdot n) = \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n)$$

$$\begin{array}{ll}
1 & 1 \quad m, n \in \mathbb{N} & A \\
1 & 2 \quad mn \in \mathbb{N} & Th. 10.4.7.3(1) \\
1 & 3 \quad \iota_{\mathbb{Z}}(mn) = [mn, 0]_{\mathbb{Z}} & Th. 17.2.3.2(2) \\
1 & 4 \quad \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n) = [mn, 0]_{\mathbb{Z}} & Th. 17.2.6.2(i)(1) \\
1 & 5 \quad \iota_{\mathbb{Z}}(mn) = \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n) & Th. 2.9.1.3(3, 4)
\end{array}$$

□

Kapitel 3

Nachfolger, Vorgänger und Normalformen

3.1 Nachfolger und Vorgänger

Definition 17.3.1.1 (Nachfolger einer ganzen Zahl).

Mengen: $\mathbb{Z}$
Funktionensymbol: $S_{\mathbb{Z}}$
Neue Symbole: $S_{\mathbb{Z}}$

$$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}, \\ \forall z \in \mathbb{Z} (S_{\mathbb{Z}}(z) := z + 1_{\mathbb{Z}})$$

- | | | |
|---|--|-----------------------------------|
| 1 | $z \in \mathbb{Z}$ | A |
| 2 | $1_{\mathbb{Z}} \in \mathbb{Z}$ | $Th. 17.2.3.6$ |
| 1 | $z + 1_{\mathbb{Z}} \in \mathbb{Z}$ | $Th. 17.2.4.7(1, 2)$ |
| 4 | $\forall z \in \mathbb{Z} (z + 1_{\mathbb{Z}} \in \mathbb{Z})$ | $\forall I (\rightarrow I(1, 3))$ |

Definition 17.3.1.2 (Vorgänger einer ganzen Zahl).

Mengen: $\mathbb{Z}$
Funktionensymbol: $P_{\mathbb{Z}}$
Neue Symbole: $P_{\mathbb{Z}}$

$$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}, \\ \forall z \in \mathbb{Z} (P_{\mathbb{Z}}(z) := z + (-1_{\mathbb{Z}}))$$

- | | | |
|---|----------------------------------|-------------------|
| 1 | $z \in \mathbb{Z}$ | A |
| 2 | $1_{\mathbb{Z}} \in \mathbb{Z}$ | $Th. 17.2.3.6$ |
| 3 | $-1_{\mathbb{Z}} \in \mathbb{Z}$ | $Th. 17.2.4.3(2)$ |

$$\begin{array}{ll} 1 & 4 \quad z + (-1_{\mathbb{Z}}) \in \mathbb{Z} \quad \text{Th. 17.2.4.7(1,3)} \\ 5 & \forall z \in \mathbb{Z} (z + (-1_{\mathbb{Z}}) \in \mathbb{Z}) \quad \forall I(\rightarrow I(1,4)) \end{array}$$

Theorem 17.3.1.1 (Nachfolgerfunktion auf den ganzen Zahlen).

Funktion: $S_{\mathbb{Z}}$

$$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$1 \quad S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \quad \text{Def. 17.3.1.1}$$

Theorem 17.3.1.2 (Vorgängerfunktion auf den ganzen Zahlen).

Funktion: $P_{\mathbb{Z}}$

$$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$1 \quad P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \quad \text{Def. 17.3.1.2}$$

Theorem 17.3.1.3 (Grundgleichung des ganzzahligen Nachfolgers).

Mengen: z

Funktion: $S_{\mathbb{Z}}$

$$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(z) = z + 1_{\mathbb{Z}}$$

$$\begin{array}{ll} 1 & 1 \quad z \in \mathbb{Z} \quad A \\ 2 & \forall u \in \mathbb{Z} (S_{\mathbb{Z}}(u) = u + 1_{\mathbb{Z}}) \quad \text{Def. 17.3.1.1} \\ 1 & 3 \quad S_{\mathbb{Z}}(z) = z + 1_{\mathbb{Z}} \quad \rightarrow E(\forall E(2), 1) \end{array}$$

Theorem 17.3.1.4 (Grundgleichung des ganzzahligen Vorgängers).

Mengen: z

Funktion: $P_{\mathbb{Z}}$

$$z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(z) = z + (-1_{\mathbb{Z}})$$

$$\begin{array}{ll} 1 & 1 \quad z \in \mathbb{Z} \quad A \\ 2 & \forall u \in \mathbb{Z} (P_{\mathbb{Z}}(u) = u + (-1_{\mathbb{Z}})) \quad \text{Def. 17.3.1.2} \\ 1 & 3 \quad P_{\mathbb{Z}}(z) = z + (-1_{\mathbb{Z}}) \quad \rightarrow E(\forall E(2), 1) \end{array}$$

Theorem 17.3.1.5 (Nachfolger auf Klassen).

Mengen: a, b

$$a, b \in \mathbb{N} \vdash S_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [\text{succ}(a), b]_{\mathbb{Z}}$$

1 1	$a, b \in \mathbb{N}$	A
1 2	$[a, b]_{\mathbb{Z}} \in \mathbb{Z}$	$Th. 17.2.2.1(1)$
3	$\forall z \in \mathbb{Z} (S_{\mathbb{Z}}(z) = z + 1_{\mathbb{Z}})$	$Def. 17.3.1.1$
1 4	$S_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} + 1_{\mathbb{Z}}$	$\rightarrow E(\forall E(3), 2)$
5	$1_{\mathbb{Z}} = [1, 0]_{\mathbb{Z}}$	$Def. 17.2.3.3(Th. 17.2.3.2(Th. 10.4.4.11))$
1 6	$[a, b]_{\mathbb{Z}} + 1_{\mathbb{Z}} = [a + 1, b + 0]_{\mathbb{Z}}$	$Def. 17.2.4.2(1, 5)$
1 7	$a + 1 = \text{succ}(a)$	$Th. 10.4.4.12(1)$
1 8	$b + 0 = b$	$Th. 10.4.4.2(1)$
1 9	$S_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [\text{succ}(a), b]_{\mathbb{Z}}$	$= E(4, 6, 7, 8)$

Theorem 17.3.1.6 (Vorgänger auf Klassen).

Mengen: a, b

$$a, b \in \mathbb{N} \vdash P_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [a, \text{succ}(b)]_{\mathbb{Z}}$$

1 1	$a, b \in \mathbb{N}$	A
1 2	$[a, b]_{\mathbb{Z}} \in \mathbb{Z}$	$Th. 17.2.2.1(1)$
3	$\forall z \in \mathbb{Z} (P_{\mathbb{Z}}(z) = z + (-1_{\mathbb{Z}}))$	$Def. 17.3.1.2$
1 4	$P_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} + (-1_{\mathbb{Z}})$	$\rightarrow E(\forall E(3), 2)$
5	$-1_{\mathbb{Z}} = [0, 1]_{\mathbb{Z}}$	$Def. 17.2.4.1(Def. 17.2.3.3, Th. 17.2.3.2)$
1 6	$[a, b]_{\mathbb{Z}} + (-1_{\mathbb{Z}}) = [a + 0, b + 1]_{\mathbb{Z}}$	$Def. 17.2.4.2(1, 5)$
1 7	$a + 0 = a$	$Th. 10.4.4.2(1)$
1 8	$b + 1 = \text{succ}(b)$	$Th. 10.4.4.12(1)$
1 9	$P_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [a, \text{succ}(b)]_{\mathbb{Z}}$	$= E(4, 6, 7, 8)$

Theorem 17.3.1.7 (Vorgänger nach Nachfolger).

Mengen: z

$$z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$$

Beweis.

Auflösen der Definitionen

Th. 17.3.1.7(i)

$$z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = (z + 1_{\mathbb{Z}}) + (-1_{\mathbb{Z}})$$

1	$1 \ z \in \mathbb{Z}$	A
1	$2 \ S_{\mathbb{Z}}(z) = z + 1_{\mathbb{Z}}$	Th. 17.3.1.3(1)
1	$3 \ S_{\mathbb{Z}}(z) \in \mathbb{Z}$	Th. 5.2.3.8(Th. 17.3.1.1,1)
1	$4 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = S_{\mathbb{Z}}(z) + (-1_{\mathbb{Z}})$	Th. 17.3.1.4(3)
1	$5 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = (z + 1_{\mathbb{Z}}) + (-1_{\mathbb{Z}}) = E(2, 4)$	

Schluss:

1	$1 \ z \in \mathbb{Z}$	A
1	$2 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = (z + 1_{\mathbb{Z}}) + (-1_{\mathbb{Z}})$	Th. 17.3.1.7(i)(1)
	$3 \ 1_{\mathbb{Z}} \in \mathbb{Z}$	Th. 17.2.3.6
	$4 \ 1_{\mathbb{Z}} + (-1_{\mathbb{Z}}) = 0_{\mathbb{Z}}$	$\wedge E1(Th. 17.2.4.11(3))$
1	$5 \ (z + 1_{\mathbb{Z}}) + (-1_{\mathbb{Z}}) = z + (1_{\mathbb{Z}} + (-1_{\mathbb{Z}}))$	Th. 17.2.4.8(1,3, Th. 17.2.4.3(3))
1	$6 \ z + (1_{\mathbb{Z}} + (-1_{\mathbb{Z}})) = z + 0_{\mathbb{Z}}$	$= E(4)$
1	$7 \ z + 0_{\mathbb{Z}} = z$	$\wedge E1(Th. 17.2.4.10(1))$
1	$8 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$	$= E(2, 5, 6, 7)$

□

Theorem 17.3.1.8 (Nachfolger nach Vorgänger).

Mengen: z

$$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$$

Beweis.

Auflösen der Definitionen

Th. 17.3.1.8(i)

$$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = (z + (-1_{\mathbb{Z}})) + 1_{\mathbb{Z}}$$

1	$1 \ z \in \mathbb{Z}$	A
1	$2 \ P_{\mathbb{Z}}(z) = z + (-1_{\mathbb{Z}})$	Th. 17.3.1.4(1)
1	$3 \ P_{\mathbb{Z}}(z) \in \mathbb{Z}$	Th. 5.2.3.8(Th. 17.3.1.2,1)
1	$4 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = P_{\mathbb{Z}}(z) + 1_{\mathbb{Z}}$	Th. 17.3.1.3(3)
1	$5 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = (z + (-1_{\mathbb{Z}})) + 1_{\mathbb{Z}} = E(2, 4)$	

Assoziativität und Inverses

Th. 17.3.1.8(ii)

$$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$$

1	$1 \ z \in \mathbb{Z}$	A
1	$2 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = (z + (-1_{\mathbb{Z}})) + 1_{\mathbb{Z}}$	Th. 17.3.1.8(i)(1)
	$3 \ 1_{\mathbb{Z}} \in \mathbb{Z}$	Th. 17.2.3.6
	$4 \ -1_{\mathbb{Z}} \in \mathbb{Z}$	Th. 17.2.4.3(3)
	$5 \ (-1_{\mathbb{Z}}) + 1_{\mathbb{Z}} = 0_{\mathbb{Z}}$	$\wedge E2(Th. 17.2.4.11(3))$
1	$6 \ (z + (-1_{\mathbb{Z}})) + 1_{\mathbb{Z}} = z + ((-1_{\mathbb{Z}}) + 1_{\mathbb{Z}})$	Th. 17.2.4.8(1,4,3)
1	$7 \ z + ((-1_{\mathbb{Z}}) + 1_{\mathbb{Z}}) = z + 0_{\mathbb{Z}}$	$= E(5)$
1	$8 \ z + 0_{\mathbb{Z}} = z$	$\wedge E1(Th. 17.2.4.10(1))$
1	$9 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$	$= E(2, 6, 7, 8)$

□

Theorem 17.3.1.9 (Bijektivität des Ganzzahlnachfolgers).

Funktionen: $S_{\mathbb{Z}}, P_{\mathbb{Z}}$

$$S_{\mathbb{Z}}: \mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$$

Beweis.

Injektivität

Th. 17.3.1.9(i)

$$x, y \in \mathbb{Z}, S_{\mathbb{Z}}(x) = S_{\mathbb{Z}}(y) \vdash x = y$$

1	$1 \ x, y \in \mathbb{Z}$	A
2	$2 \ S_{\mathbb{Z}}(x) = S_{\mathbb{Z}}(y)$	A
1	$3 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(x)) = x$	Th. 17.3.1.7(1)
1	$4 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(y)) = y$	Th. 17.3.1.7(1)
2	$5 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(x)) = P_{\mathbb{Z}}(S_{\mathbb{Z}}(y))$	$= E(2)$
1,2	$6 \ x = y$	$= E(3, 4, 5)$

Surjektivität

Th. 17.3.1.9(ii)

$$y \in \mathbb{Z} \vdash \exists x \in \mathbb{Z} S_{\mathbb{Z}}(x) = y$$

1	$1 \ y \in \mathbb{Z}$	A
1	$2 \ P_{\mathbb{Z}}(y) \in \mathbb{Z}$	Th. 5.2.3.8(Th. 17.3.1.2,1)
1	$3 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(y)) = y$	Th. 17.3.1.8(1)
1	$4 \ \exists x \in \mathbb{Z} S_{\mathbb{Z}}(x) = y$	$\exists I(\wedge I(2, 3))$

Schluss:

$$1 \ S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \quad \text{Th. 17.3.1.1}$$

2	$\forall x, y \in \mathbb{Z} (S_{\mathbb{Z}}(x) = S_{\mathbb{Z}}(y) \rightarrow x = y)$	Th. 17.3.1.9(i)
3	$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	Def. 6.2.1.1(1,2)
4	$\forall y \in \mathbb{Z} \exists x \in \mathbb{Z} S_{\mathbb{Z}}(x) = y$	Th. 17.3.1.9(ii)
5	$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	Def. 7.2.1.1(1,4)
6	$S_{\mathbb{Z}}: \mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$	Th. 8.2.1.3(3,5)

□

3.2 Normalformen

Theorem 17.3.2.1 (Positiver Strahl und Nachfolger).

Mengen: n

$$n \in \mathbb{N} \vdash S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)) = \iota_{\mathbb{Z}}(\text{succ}(n))$$

1	1	$n \in \mathbb{N}$	A
1	2	$S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)) = \iota_{\mathbb{Z}}(n) + 1_{\mathbb{Z}}$	Th. 17.3.1.3(Th. 5.2.3.8(Th. 17.2.3.1, 1))
	3	$1_{\mathbb{Z}} = \iota_{\mathbb{Z}}(1)$	Def. 17.2.3.3
1	4	$\iota_{\mathbb{Z}}(n) + \iota_{\mathbb{Z}}(1) = \iota_{\mathbb{Z}}(n + 1)$	Th. 2.9.1.1(Th. 17.2.6.1(1, Th. 10.4.4.11))
1	5	$n + 1 = \text{succ}(n)$	Th. 10.4.4.12(1)
1	6	$S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)) = \iota_{\mathbb{Z}}(\text{succ}(n))$	$= E(2, 3, 4, 5)$

Theorem 17.3.2.2 (Negativer Strahl und Vorgänger).

Mengen: n

$$n \in \mathbb{N} \vdash P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n)) = -\iota_{\mathbb{Z}}(\text{succ}(n))$$

1	1	$n \in \mathbb{N}$	A
1	2	$\iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}$	Th. 17.2.3.2(1)
1	3	$-\iota_{\mathbb{Z}}(n) = [0, n]_{\mathbb{Z}}$	Def. 17.2.4.1(1, 2)
1	4	$P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n)) = [0, \text{succ}(n)]_{\mathbb{Z}}$	Th. 17.3.1.6(1, 3)
1	5	$\iota_{\mathbb{Z}}(\text{succ}(n)) = [\text{succ}(n), 0]_{\mathbb{Z}}$	Th. 17.2.3.2(Ax. 10.3.4(1))
1	6	$-\iota_{\mathbb{Z}}(\text{succ}(n)) = [0, \text{succ}(n)]_{\mathbb{Z}}$	Def. 17.2.4.1(1, 5)
1	7	$P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n)) = -\iota_{\mathbb{Z}}(\text{succ}(n))$	Th. 2.9.1.3(4, 6)

Theorem 17.3.2.3 (Normalform ganzer Zahlen).

Mengen: z, a, b, n

$$z \in \mathbb{Z} \vdash \exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n))$$

Beweis.

Fall $b \leq a$

Th. 17.3.2.3(i)

$$\begin{aligned} a, b \in \mathbb{N}, \quad b \leq a &\vdash \\ \exists n \in \mathbb{N} \quad [a, b]_{\mathbb{Z}} = \iota_{\mathbb{Z}}(n) \end{aligned}$$

$$\begin{array}{ll} 1 & 1 \ a, b \in \mathbb{N} \quad A \\ 2 & 2 \ b \leq a \quad A \\ 1,2 & 3 \ \exists n \in \mathbb{N} (b + n = a) \quad \text{Th. 10.4.5.6(1,2)} \\ 4 & 4 \ n \in \mathbb{N} \wedge b + n = a \quad A \\ 1,4 & 5 \ a + 0 = b + n \quad = E(\text{Th. 10.4.4.2(1)}, \wedge E2(4)) \\ 1,4 & 6 \ [a, b]_{\mathbb{Z}} = [n, 0]_{\mathbb{Z}} \quad \text{Th. 17.2.2.2(1,4,5)} \\ 4 & 7 \ \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}} \quad \text{Th. 17.2.3.2(4)} \\ 1,4 & 8 \ [a, b]_{\mathbb{Z}} = \iota_{\mathbb{Z}}(n) \quad \text{Th. 2.9.1.3(6,7)} \\ 1,2 & 9 \ \exists n \in \mathbb{N} \ [a, b]_{\mathbb{Z}} = \iota_{\mathbb{Z}}(n) \exists E(3, 4, 8) \end{array}$$

Fall $a \leq b$

Th. 17.3.2.3(ii)

$$\begin{aligned} a, b \in \mathbb{N}, \quad a \leq b &\vdash \\ \exists n \in \mathbb{N} \quad [a, b]_{\mathbb{Z}} = -\iota_{\mathbb{Z}}(n) \end{aligned}$$

$$\begin{array}{ll} 1 & 1 \ a, b \in \mathbb{N} \quad A \\ 2 & 2 \ a \leq b \quad A \\ 1,2 & 3 \ \exists n \in \mathbb{N} (a + n = b) \quad \text{Th. 10.4.5.6(1,2)} \\ 4 & 4 \ n \in \mathbb{N} \wedge a + n = b \quad A \\ 1,4 & 5 \ a + n = b + 0 \quad = E(\wedge E2(4), \text{Th. 10.4.4.2(1)}) \\ 1,4 & 6 \ [a, b]_{\mathbb{Z}} = [0, n]_{\mathbb{Z}} \quad \text{Th. 17.2.2.2(1,4,5)} \\ 4 & 7 \ \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}} \quad \text{Th. 17.2.3.2(4)} \\ 4 & 8 \ -\iota_{\mathbb{Z}}(n) = [0, n]_{\mathbb{Z}} \quad \text{Def. 17.2.4.1(4,7)} \\ 1,4 & 9 \ [a, b]_{\mathbb{Z}} = -\iota_{\mathbb{Z}}(n) \quad \text{Th. 2.9.1.3(6,8)} \\ 1,2 & 10 \ \exists n \in \mathbb{N} \ [a, b]_{\mathbb{Z}} = -\iota_{\mathbb{Z}}(n) \exists E(3, 4, 9) \end{array}$$

Schluss:

$$\begin{array}{ll} 1 & 1 \ z \in \mathbb{Z} \quad A \\ 1 & 2 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}} \quad \text{Th. 17.2.2.3(1)} \\ 3 & 3 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \quad A \\ 3 & 4 \ b \leq a \vee a \leq b \quad \text{Th. 10.4.5.22(3)} \\ 5 & 5 \ b \leq a \quad A \\ 3,5 & 6 \ \exists n \in \mathbb{N} \ [a, b]_{\mathbb{Z}} = \iota_{\mathbb{Z}}(n) \quad \text{Th. 17.3.2.3(i)(3,5)} \\ 3,5 & 7 \ \exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) = E(3, 6) \\ 8 & 8 \ a \leq b \quad A \end{array}$$

$$\begin{array}{ll}
3,8 & 9 \exists n \in \mathbb{N} [a, b]_{\mathbb{Z}} = -\iota_{\mathbb{Z}}(n) \quad \text{Th. 17.3.2.3(ii)(3,8)} \\
3,8 & 10 \exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) = E(3, 9) \\
3 & 11 \exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) \vee E(4, 5, 7, 8, 10) \\
1 & 12 \exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) \exists E(2, 3, 11)
\end{array}$$

□

Kapitel 4

Ordnung der ganzen Zahlen

4.1 Translationsinvarianz der natürlichen Ordnung

Theorem 17.4.1.1 (Translationsinvarianz der natürlichen Ordnung).

Mengen: a, b, c, k

$$a, b, c \in \mathbb{N} \vdash a \leq b \leftrightarrow a + c \leq b + c$$

Beweis.

Erhaltung unter Translation

Th. 17.4.1.1(i)

$$a, b, c \in \mathbb{N}, a \leq b \vdash a + c \leq b + c$$

1	$1 \ a, b, c \in \mathbb{N}$	A
2	$2 \ a \leq b$	A
1,2	$3 \ \exists k \in \mathbb{N} (a + k = b)$	Th. 10.4.5.6(1,2)
4	$4 \ k \in \mathbb{N} \wedge a + k = b$	A
1,4	$5 \ (a + c) + k = (a + k) + c$	Th. 10.4.4.10(1)
1,4	$6 \ (a + k) + c = b + c$	$= E(\wedge E2(4))$
1,4	$7 \ (a + c) + k = b + c$	Th. 2.9.1.2(5,6)
1,4	$8 \ a + c \leq b + c$	Th. 10.4.5.6(1, $\exists I(\wedge I(\wedge E1(4), 7))$)
1,2	$9 \ a + c \leq b + c$	$\exists E(3, 4, 8)$

Kürzen einer Translation

Th. 17.4.1.1(ii)

$$a, b, c \in \mathbb{N}, a + c \leq b + c \vdash a \leq b$$

1	$1 \ a, b, c \in \mathbb{N}$	A
---	------------------------------	-----

2	$2 a + c \leq b + c$	A
1,2	$3 \exists k \in \mathbb{N} ((a + c) + k = b + c)$	Th. 10.4.5.6(1,2)
4	$4 k \in \mathbb{N} \wedge (a + c) + k = b + c$	A
1,4	$5 (a + k) + c = (a + c) + k$	Th. 10.4.4.10(1)
1,4	$6 (a + c) + k = b + c$	$\wedge E2(4)$
1,4	$7 (a + k) + c = b + c$	Th. 2.9.1.2(5,6)
1,4	$8 a + k = b$	Th. 10.4.5.30(Th. 10.4.4.1(1,4),1,1,7)
1,4	$9 a \leq b$	Th. 10.4.5.6(1, $\exists I(\wedge I(\wedge E1(4), 8))$)
1,2	$10 a \leq b$	$\exists E(3, 4, 9)$

Schluss:

1	$1 a, b, c \in \mathbb{N}$	A
1	$2 a \leq b \rightarrow a + c \leq b + c$	Th. 17.4.1.1(i)(1)
1	$3 a + c \leq b + c \rightarrow a \leq b$	Th. 17.4.1.1(ii)(1)
1	$4 a \leq b \leftrightarrow a + c \leq b + c \leftrightarrow I(2, 3)$	

□

Theorem 17.4.1.2 (Addition natürlicher Ungleichungen).

Mengen: a, b, c, d

$$a, b, c, d \in \mathbb{N}, a \leq b, c \leq d \vdash a + c \leq b + d$$

Beweis.

Translation der ersten Ungleichung

Th. 17.4.1.2(i)

$$a, b, c \in \mathbb{N}, a \leq b \vdash a + c \leq b + c$$

1	$1 a, b, c \in \mathbb{N}$	A
2	$2 a \leq b$	A
1,2	$3 a + c \leq b + c$	Th. 17.4.1.1(i)(1,2)

Translation der zweiten Ungleichung

Th. 17.4.1.2(ii)

$$b, c, d \in \mathbb{N}, c \leq d \vdash b + c \leq b + d$$

1	$1 b, c, d \in \mathbb{N}$	A
2	$2 c \leq d$	A
1,2	$3 c + b \leq d + b$	Th. 17.4.1.1(i)(1,2)
1	$4 b + c = c + b$	Th. 10.4.4.8(1)

$$\begin{array}{l} 1 \quad 5 \quad d + b = b + d \text{ Th. 10.4.4.8(1)} \\ 1,2 \quad 6 \quad b + c \leq b + d = E(4, 3, 5) \end{array}$$

Verknüpfung durch Transitivität

Th. 17.4.1.2(iii)

$$a, b, c, d \in \mathbb{N}, \quad a \leq b, \quad c \leq d \vdash a + c \leq b + d$$

$$\begin{array}{l} 1 \quad 1 \quad a, b, c, d \in \mathbb{N} \quad A \\ 2 \quad 2 \quad a \leq b \quad A \\ 3 \quad 3 \quad c \leq d \quad A \\ 1,2 \quad 4 \quad a + c \leq b + c \text{ Th. 17.4.1.2(i)(1,2)} \\ 1,3 \quad 5 \quad b + c \leq b + d \text{ Th. 17.4.1.2(ii)(1,3)} \\ 1 \quad 6 \quad a + c \in \mathbb{N} \quad \text{Th. 10.4.4.1(1)} \\ 1 \quad 7 \quad b + c \in \mathbb{N} \quad \text{Th. 10.4.4.1(1)} \\ 1 \quad 8 \quad b + d \in \mathbb{N} \quad \text{Th. 10.4.4.1(1)} \\ 1,2,3 \quad 9 \quad a + c \leq b + d \text{ Th. 10.4.5.24(6,7,8,4,5)} \end{array}$$

□

4.2 Quotientenordnung

Theorem 17.4.2.1 (Repräsentantenunabhängigkeit der Ganzzahlordnung).

Mengen: $a, b, a', b', c, d, c', d'$

$$\begin{array}{l} a, b, a', b', c, d, c', d' \in \mathbb{N}, \quad [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}, \quad [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\ (a + d \leq c + b) \leftrightarrow (a' + d' \leq c' + b') \end{array}$$

Beweis.

Hinrichtung

Th. 17.4.2.1(i)

$$\begin{array}{l} a, b, a', b', c, d, c', d' \in \mathbb{N}, \quad a + b' = b + a', \quad c + d' = d + c', \quad a + d \leq c + b \vdash \\ a' + d' \leq c' + b' \end{array}$$

$$\begin{array}{ll} 1 \quad 1 \quad a, b, a', b', c, d, c', d' \in \mathbb{N} & A \\ 2 \quad 2 \quad a + b' = b + a' & A \\ 3 \quad 3 \quad c + d' = d + c' & A \\ 4 \quad 4 \quad a + d \leq c + b & A \end{array}$$

$$\begin{array}{ll}
1,4 & 5 \ (a+d) + (b'+c') \leq (c+b) + (b'+c') \text{ Th. 17.4.1.1(i)(1,4)} \\
1,2,3 & 6 \ (a+d) + (b'+c') = (a'+d') + (b+c) \text{ Th. 10.4.4.10(1,2,3)} \\
1 & 7 \ (c+b) + (b'+c') = (c'+b') + (b+c) \text{ Th. 10.4.4.10(1)} \\
1,2,3,4 & 8 \ (a'+d') + (b+c) \leq (c'+b') + (b+c) = E(5,6,7) \\
1,2,3,4 & 9 \ a' + d' \leq c' + b' \text{ Th. 17.4.1.1(ii)(1,8)}
\end{array}$$

Rückrichtung

Th. 17.4.2.1(ii)

$$\begin{array}{l}
a, b, a', b', c, d, c', d' \in \mathbb{N}, \ a + b' = b + a', \ c + d' = d + c', \ a' + d' \leq c' + b' \vdash \\
a + d \leq c + b
\end{array}$$

$$\begin{array}{ll}
1 & 1 \ a, b, a', b', c, d, c', d' \in \mathbb{N} \quad A \\
2 & 2 \ a + b' = b + a' \quad A \\
3 & 3 \ c + d' = d + c' \quad A \\
4 & 4 \ a' + d' \leq c' + b' \quad A \\
1,2 & 5 \ a' + b = b' + a \quad \text{Th. 2.9.1.1(Th. 10.4.4.8(2))} \\
1,3 & 6 \ c' + d = d' + c \quad \text{Th. 2.9.1.1(Th. 10.4.4.8(3))} \\
1,2,3,4 & 7 \ a + d \leq c + b \quad \text{Th. 17.4.2.1(i)(1,5,6,4)}
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 & 1 \ a, b, a', b', c, d, c', d' \in \mathbb{N} \quad A \\
2 & 2 \ [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}} \quad A \\
3 & 3 \ [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \quad A \\
1,2 & 4 \ a + b' = b + a' \quad \text{Th. 17.2.2.2(1,2)} \\
1,3 & 5 \ c + d' = d + c' \quad \text{Th. 17.2.2.2(1,3)} \\
1,2,3 & 6 \ a + d \leq c + b \rightarrow a' + d' \leq c' + b' \quad \text{Th. 17.4.2.1(i)(1,4,5)} \\
1,2,3 & 7 \ a' + d' \leq c' + b' \rightarrow a + d \leq c + b \quad \text{Th. 17.4.2.1(ii)(1,4,5)} \\
1,2,3 & 8 \ (a + d \leq c + b) \leftrightarrow (a' + d' \leq c' + b') \leftrightarrow I(6,7)
\end{array}$$

□

Definition 17.4.2.1 (Ordnungsoperator der ganzen Zahlen).

Mengen: a, b, c, d
Relationsoperatoren: $\leq_{\mathbb{Z}}$
Neue Symbole: $\leq_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} \leq_{\mathbb{Z}} [c, d]_{\mathbb{Z}} := a + d \leq c + b$$

Definition 17.4.2.2 (Notation der Ganzzahlordnung).

Mengen: z, w
Relationsoperatoren: $\leq, \leq_{\mathbb{Z}}$

$$z, w \in \mathbb{Z} \vdash z \leq w := z \leq_{\mathbb{Z}} w$$

Theorem 17.4.2.2 (Klassencharakterisierung der Ganzzahlordnung).

Mengen: a, b, c, d
Relationsoperatoren: $\leq, \leq_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} \leq [c, d]_{\mathbb{Z}} \leftrightarrow a + d \leq c + b$$

1	1	$a, b, c, d \in \mathbb{N}$	A
1	2	$[a, b]_{\mathbb{Z}} \in \mathbb{Z}$	Th. 17.2.2.1(1)
1	3	$[c, d]_{\mathbb{Z}} \in \mathbb{Z}$	Th. 17.2.2.1(1)
1	4	$[a, b]_{\mathbb{Z}} \leq [c, d]_{\mathbb{Z}} \leftrightarrow [a, b]_{\mathbb{Z}} \leq_{\mathbb{Z}} [c, d]_{\mathbb{Z}}$	Def. 17.4.2.2(2, 3)
1	5	$[a, b]_{\mathbb{Z}} \leq_{\mathbb{Z}} [c, d]_{\mathbb{Z}} \leftrightarrow a + d \leq c + b$	Def. 17.4.2.1(1)
1	6	$[a, b]_{\mathbb{Z}} \leq [c, d]_{\mathbb{Z}} \leftrightarrow a + d \leq c + b$	Th. 2.2.3.5(4, 5)

Theorem 17.4.2.3 (Totale Ordnung der ganzen Zahlen).

Mengen: z, w, u
Relationsoperatoren: $\leq_{\mathbb{Z}}$

$$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$$

Beweis.

Reflexivität Th. 17.4.2.3(i)

$$z \in \mathbb{Z} \vdash z \leq z$$

1	1	$z \in \mathbb{Z}$	A
1	2	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(1)
3	3	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3	4	$a + b \leq a + b$	Th. 10.4.5.10(Th. 10.4.4.1(3))
3	5	$[a, b]_{\mathbb{Z}} \leq [a, b]_{\mathbb{Z}}$	Th. 17.4.2.2(3, 4)
3	6	$z \leq z$	$= E(3, 5)$
1	7	$z \leq z$	$\exists E(2, 3, 6)$

Antisymmetrie Th. 17.4.2.3(ii)

$$z, w \in \mathbb{Z}, z \leq w, w \leq z \vdash z = w$$

1	1	$z, w \in \mathbb{Z}$	A
2	2	$z \leq w$	A
3	3	$w \leq z$	A
1	4	$z \in \mathbb{Z}$	$\wedge E1(1)$

1	5	$w \in \mathbb{Z}$	$\wedge E2(1)$
1	6	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(4)
1	7	$\exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 17.2.2.3(5)
8	8	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A$	
2,8	9	$a + d \leq c + b$	Th. 17.4.2.2(8,2)
3,8	10	$c + b \leq a + d$	Th. 17.4.2.2(8,3)
8	11	$a + d \in \mathbb{N}$	Th. 10.4.4.1(8)
8	12	$c + b \in \mathbb{N}$	Th. 10.4.4.1(8)
2,3,8	13	$a + d = c + b$	Th. 10.4.5.25(11,12,9,10)
2,3,8	14	$[a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}}$	Th. 17.2.2.2(8,13)
2,3,8	15	$z = w$	$= E(8, 14)$
1,2,3	16	$z = w$	$\exists E(6, 7, 8, 15)$

Transitivität

Th. 17.4.2.3(iii)

$$z, w, u \in \mathbb{Z}, z \leq w, w \leq u \vdash z \leq u$$

1	1	$z, w, u \in \mathbb{Z}$	A
2	2	$z \leq w$	A
3	3	$w \leq u$	A
1	4	$z \in \mathbb{Z}$	$\wedge E1(1)$
1	5	$w \in \mathbb{Z}$	$\wedge E1(\wedge E2(1))$
1	6	$u \in \mathbb{Z}$	$\wedge E2(\wedge E2(1))$
4	7	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(4)
5	8	$\exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 17.2.2.3(5)
6	9	$\exists e, f \in \mathbb{N} u = [e, f]_{\mathbb{Z}}$	Th. 17.2.2.3(6)
10	10	$a, b, c, d, e, f \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [e, f]_{\mathbb{Z}} A$	
2,10	11	$a + d \leq c + b$	Th. 17.4.2.2(10,2)
3,10	12	$c + f \leq e + d$	Th. 17.4.2.2(10,3)
2,3,10	13	$(a + d) + (c + f) \leq (c + b) + (e + d)$	Th. 17.4.1.2(10,11,12)
10	14	$(a + d) + (c + f) = (a + f) + (c + d)$	Th. 10.4.4.10(10)
10	15	$(c + b) + (e + d) = (b + e) + (c + d)$	Th. 10.4.4.10(10)
2,3,10	16	$(a + f) + (c + d) \leq (b + e) + (c + d)$	$= E(13, 14, 15)$
2,3,10	17	$a + f \leq b + e$	Th. 17.4.1.1(ii)(10,16)
2,3,10	18	$z \leq u$	Th. 17.4.2.2(10,17)
1,2,3	19	$z \leq u$	$\exists E(7, 8, 9, 10, 18)$

Totalität

Th. 17.4.2.3(iv)

$$z, w \in \mathbb{Z} \vdash z \leq w \vee w \leq z$$

1	1	$z, w \in \mathbb{Z}$	A
---	---	-----------------------	-----

1	$2 \ z \in \mathbb{Z}$	$\wedge E1(1)$
1	$3 \ w \in \mathbb{Z}$	$\wedge E2(1)$
2	$4 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(2)
3	$5 \ \exists c, d \in \mathbb{N} \ w = [c, d]_{\mathbb{Z}}$	Th. 17.2.2.3(3)
6	$6 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A$	
6	$7 \ a + d \leq c + b \vee c + b \leq a + d$	Th. 10.4.5.22(Th. 10.4.4.1(6), Th. 10.4.4.1(6))
6	$8 \ z \leq w \vee w \leq z$	Th. 17.4.2.2(6,7)
1	$9 \ z \leq w \vee w \leq z$	$\exists E(4, 5, 6, 8)$

Schluss:

	$1 \ \forall z \in \mathbb{Z} (z \leq z)$	Th. 17.4.2.3(i)
2	$2 \ z \in \mathbb{Z}$	A
3	$3 \ w \in \mathbb{Z}$	A
4	$4 \ u \in \mathbb{Z}$	A
5	$5 \ z \leq w$	A
6	$6 \ w \leq u$	A
2,3,4,5,6	$7 \ z \leq u$	Th. 17.4.2.3(iii)(2,3,4,5,6)
8	$8 \ w \leq z$	A
2,3,5,8	$9 \ z = w$	Th. 17.4.2.3(ii)(2,3,5,8)
10	$\text{PartOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$	Def. 12.2.1.1(1,7,9)
2,3	$11 \ z \leq w \vee w \leq z$	Th. 17.4.2.3(iv)(2,3)
12	$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$	Def. 15.2.1.1(10,11)

□

Definition 17.4.2.3 (Strikte Ordnung der ganzen Zahlen).

Mengen: z, w

Relationsoperatoren: $<$

$$z, w \in \mathbb{Z} \vdash z < w := z \leq w \wedge z \neq w$$

Theorem 17.4.2.4 (Jede ganze Zahl ist kleiner als ihr Nachfolger).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash z < S_{\mathbb{Z}}(z)$$

Beweis.

Ordnungsteil

Th. 17.4.2.4(i)

$$z \in \mathbb{Z} \vdash z \leq S_{\mathbb{Z}}(z)$$

1	$z \in \mathbb{Z}$	A
1	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(1)
3	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3	$S_{\mathbb{Z}}(z) = [\text{succ}(a), b]_{\mathbb{Z}}$	Th. 17.3.1.5(3)
3	$a \leq \text{succ}(a)$	Th. 10.4.5.12(3)
3	$a + b \leq \text{succ}(a) + b$	Th. 17.4.1.1(i)(3,5)
3	$z \leq S_{\mathbb{Z}}(z)$	Th. 17.4.2.2(3,4,6)
1	$z \leq S_{\mathbb{Z}}(z)$	$\exists E(2, 3, 7)$

Ungleichheitsteil

Th. 17.4.2.4(ii)

$$z \in \mathbb{Z} \vdash z \neq S_{\mathbb{Z}}(z)$$

1	$z \in \mathbb{Z}$	A
2	$z = S_{\mathbb{Z}}(z)$	A
1	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 17.2.2.3(1)
4	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
4	$S_{\mathbb{Z}}(z) = [\text{succ}(a), b]_{\mathbb{Z}}$	Th. 17.3.1.5(4)
2,4	$[a, b]_{\mathbb{Z}} = [\text{succ}(a), b]_{\mathbb{Z}}$	$= E(4, 5, 2)$
2,4	$a + b = b + \text{succ}(a)$	Th. 17.2.2.2(4,6)
4	$b + \text{succ}(a) = \text{succ}(a) + b$	Th. 10.4.4.8(4)
2,4	$a + b = \text{succ}(a) + b$	$= E(7, 8)$
2,4	$a = \text{succ}(a)$	Th. 10.4.5.30(4,9)
4	$a < \text{succ}(a)$	Th. 10.4.5.13(4)
2,4	$\perp$	Th. 10.4.5.28(4,11,10)
1,2	$\perp$	$\exists E(3, 4, 12)$
1	$z \neq S_{\mathbb{Z}}(z)$	$\neg I(2, 13)$

Schluss:

1	$z \in \mathbb{Z}$	A
1	$z \leq S_{\mathbb{Z}}(z)$	Th. 17.4.2.4(i)(1)
1	$z \neq S_{\mathbb{Z}}(z)$	Th. 17.4.2.4(ii)(1)
1	$z < S_{\mathbb{Z}}(z)$	Def. 17.4.2.3(1, $\wedge I(2, 3)$)

□

Kapitel 5

Erweiterte Peano-Axiome der ganzen Zahlen

5.1 Herleitung der zweiseitigen Induktion

Theorem 17.5.1.1 (Zweiseitige Induktion im Quotientenmodell).

Einstelliges Prädikat: P
Mengen: z, n

$$P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))), \\ \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \forall z \in \mathbb{Z} P(z)$$

Beweis.

Induktion auf dem nichtnegativen Strahl

Th. 17.5.1.1(i)

$$P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))) \vdash \\ \forall n \in \mathbb{N} P(\iota_{\mathbb{Z}}(n))$$

1	1	$P(0_{\mathbb{Z}})$	A
2	2	$\forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z)))$	A
	3	$0_{\mathbb{Z}} = \iota_{\mathbb{Z}}(0)$	Def. 17.2.3.2
1	4	$P(\iota_{\mathbb{Z}}(0))$	$= E(3, 1)$
5	5	$n \in \mathbb{N}$	A
6	6	$P(\iota_{\mathbb{Z}}(n))$	A
5	7	$\iota_{\mathbb{Z}}(n) \in \mathbb{Z}$	Th. 5.2.3.8(Th. 17.2.3.1,5)
2,5	8	$P(\iota_{\mathbb{Z}}(n)) \rightarrow P(S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)))$	$\rightarrow E(\forall E(2), 7)$
2,5,6	9	$P(S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)))$	$\rightarrow E(8, 6)$
	5	10 $S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)) = \iota_{\mathbb{Z}}(\text{succ}(n))$	Th. 17.3.2.1(5)
2,5,6	11	$P(\iota_{\mathbb{Z}}(\text{succ}(n)))$	$= E(10, 9)$

$$\begin{array}{ll}
2,5 \quad 12 \quad P(\iota_{\mathbb{Z}}(n)) \rightarrow P(\iota_{\mathbb{Z}}(\text{succ}(n))) & \rightarrow I(6, 11) \\
2 \quad 13 \quad \forall n \in \mathbb{N} (P(\iota_{\mathbb{Z}}(n)) \rightarrow P(\iota_{\mathbb{Z}}(\text{succ}(n)))) \forall I(\rightarrow I(5, 12)) & \\
1,2 \quad 14 \quad \forall n \in \mathbb{N} P(\iota_{\mathbb{Z}}(n)) & \text{Ax. 10.3.8(4,13)}
\end{array}$$

Induktion auf dem nichtpositiven Strahl

Th. 17.5.1.1(ii)

$$\begin{array}{l}
P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \\
\forall n \in \mathbb{N} P(-\iota_{\mathbb{Z}}(n))
\end{array}$$

$$\begin{array}{ll}
1 \quad 1 \quad P(0_{\mathbb{Z}}) & A \\
2 \quad 2 \quad \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) & A \\
3 \quad \iota_{\mathbb{Z}}(0) = 0_{\mathbb{Z}} & \text{Th. 2.9.1.1(Def. 17.2.3.2)} \\
4 \quad -0_{\mathbb{Z}} = 0_{\mathbb{Z}} & \text{Th. 17.2.4.12} \\
1 \quad 5 \quad P(-\iota_{\mathbb{Z}}(0)) & = E(3, 4, 1) \\
6 \quad 6 \quad n \in \mathbb{N} & A \\
7 \quad 7 \quad P(-\iota_{\mathbb{Z}}(n)) & A \\
6 \quad 8 \quad -\iota_{\mathbb{Z}}(n) \in \mathbb{Z} & \text{Th. 17.2.4.3(Th. 5.2.3.8(Th. 17.2.3.1,6))} \\
2,6 \quad 9 \quad P(-\iota_{\mathbb{Z}}(n)) \rightarrow P(P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n))) & \rightarrow E(\forall E(2), 8) \\
2,6,7 \quad 10 \quad P(P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n))) & \rightarrow E(9, 7) \\
6 \quad 11 \quad P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n)) = -\iota_{\mathbb{Z}}(\text{succ}(n)) & \text{Th. 17.3.2.2(6)} \\
2,6,7 \quad 12 \quad P(-\iota_{\mathbb{Z}}(\text{succ}(n))) & = E(11, 10) \\
2,6 \quad 13 \quad P(-\iota_{\mathbb{Z}}(n)) \rightarrow P(-\iota_{\mathbb{Z}}(\text{succ}(n))) & \rightarrow I(7, 12) \\
2 \quad 14 \quad \forall n \in \mathbb{N} (P(-\iota_{\mathbb{Z}}(n)) \rightarrow P(-\iota_{\mathbb{Z}}(\text{succ}(n)))) \forall I(\rightarrow I(6, 13)) & \\
1,2 \quad 15 \quad \forall n \in \mathbb{N} P(-\iota_{\mathbb{Z}}(n)) & \text{Ax. 10.3.8(5,14)}
\end{array}$$

Schluss mittels Normalform:

$$\begin{array}{ll}
1 \quad 1 \quad P(0_{\mathbb{Z}}) & A \\
2 \quad 2 \quad \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))) & A \\
3 \quad 3 \quad \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) & A \\
1,2 \quad 4 \quad \forall n \in \mathbb{N} P(\iota_{\mathbb{Z}}(n)) & \text{Th. 17.5.1.1(i)(1,2)} \\
1,3 \quad 5 \quad \forall n \in \mathbb{N} P(-\iota_{\mathbb{Z}}(n)) & \text{Th. 17.5.1.1(ii)(1,3)} \\
6 \quad 6 \quad z \in \mathbb{Z} & A \\
6 \quad 7 \quad \exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) & \text{Th. 17.3.2.3(6)} \\
8 \quad 8 \quad n \in \mathbb{N} \wedge (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) & A \\
4,8 \quad 9 \quad P(\iota_{\mathbb{Z}}(n)) & \rightarrow E(\forall E(4), \wedge E1(8)) \\
5,8 \quad 10 \quad P(-\iota_{\mathbb{Z}}(n)) & \rightarrow E(\forall E(5), \wedge E1(8)) \\
11 \quad 11 \quad z = \iota_{\mathbb{Z}}(n) & A \\
4,8,11 \quad 12 \quad P(z) & = E(11, 9) \\
13 \quad 13 \quad z = -\iota_{\mathbb{Z}}(n) & A \\
5,8,13 \quad 14 \quad P(z) & = E(13, 10) \\
4,5,8 \quad 15 \quad P(z) & \vee E(\wedge E2(8), 11, 12, 13, 14) \\
4,5,6 \quad 16 \quad P(z) & \exists E(7, 8, 15)
\end{array}$$

$$1,2,3 \quad 17 \forall z \in \mathbb{Z} P(z)$$

$$\forall I(\rightarrow I(6, 16))$$

□

5.2 Axiomatische Zusammenfassung

Die folgenden Axiome erweitern die Peano-Struktur um einen überall definierten Vorgänger. Anders als auf $\mathbb{N}$ ist der Nachfolger auf $\mathbb{Z}$ bijektiv. Die totale Ordnung und die strikte Nachfolgerorientierung verhindern zyklische Nachfolgermodelle; die zweiseitige Induktion verbindet den positiven und den negativen Strahl.

Metalogisch ist dabei zwischen zwei Lesarten zu unterscheiden: Als erststufiges Axiomenschema ist die Beschreibung nicht kategorisch und besitzt nichtstandardmäßige Modelle. Unter voller zweitstufiger Lesart der Induktion beschreibt sie dagegen die intendierte zweiseitige, von $0_{\mathbb{Z}}$ durch Nachfolger und Vorgänger erzeugte Kette.

Definition 17.5.2.1 (Erweiterte Peano-Axiome der ganzen Zahlen).

Axiome:

<i>Null:</i>	$0_{\mathbb{Z}} \in \mathbb{Z}$	<i>Ax. 17.5.2.1</i>
<i>Nachfolger:</i>	$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	<i>Ax. 17.5.2.2</i>
<i>Vorgänger:</i>	$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	<i>Ax. 17.5.2.3</i>
<i>Vorgänger nach Nachfolger:</i>	$z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$	<i>Ax. 17.5.2.4</i>
<i>Nachfolger nach Vorgänger:</i>	$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$	<i>Ax. 17.5.2.5</i>
<i>Totale Ordnung:</i>	$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$	<i>Ax. 17.5.2.6</i>
<i>Nachfolgerorientierung:</i>	$z \in \mathbb{Z} \vdash z < S_{\mathbb{Z}}(z)$	<i>Ax. 17.5.2.7</i>
<i>Zweiseitige Induktion:</i>	$P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))),$ $\forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \forall z \in \mathbb{Z} P(z)$	<i>Ax. 17.5.2.8</i>

Neue Symbole:

<i>Ganze Zahlen:</i>	$\mathbb{Z}$
<i>Null:</i>	$0_{\mathbb{Z}}$
<i>Nachfolger:</i>	$S_{\mathbb{Z}}$
<i>Vorgänger:</i>	$P_{\mathbb{Z}}$
<i>Ordnung:</i>	$\leq_{\mathbb{Z}}$

$$\text{ZPeano}(\mathbb{Z}, 0_{\mathbb{Z}}, S_{\mathbb{Z}}, P_{\mathbb{Z}}, \leq_{\mathbb{Z}})$$

Theorem 17.5.2.1 (Das Quotientenmodell erfüllt die Ganzzahlaxiome).

<i>Mengen:</i>	$z, D_{\mathbb{Z}}$
<i>Funktionen:</i>	$S_{\mathbb{Z}}, P_{\mathbb{Z}}$
<i>Relationsoperatoren:</i>	$\sim_{\mathbb{Z}}, \leq_{\mathbb{Z}}$

$$\mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}} \wedge$$

$$\text{ZPeano}(\mathbb{Z}, 0_{\mathbb{Z}}, S_{\mathbb{Z}}, P_{\mathbb{Z}}, \leq_{\mathbb{Z}})$$

Beweis.

Träger und Abbildungen

Th. 17.5.2.1(i)

$$0_{\mathbb{Z}} \in \mathbb{Z} \wedge S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \wedge \\ P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$\begin{array}{ll} 1 \ 0_{\mathbb{Z}} \in \mathbb{Z} & \text{Th. 17.2.3.5} \\ 2 \ S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} & \text{Th. 17.3.1.1} \\ 3 \ P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} & \text{Th. 17.3.1.2} \\ 4 \ 0_{\mathbb{Z}} \in \mathbb{Z} \wedge S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \wedge P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \wedge I(1, \wedge I(2, 3)) \end{array}$$

Gegenseitige Inversen

Th. 17.5.2.1(ii)

$$\forall z \in \mathbb{Z} \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z \wedge \\ \forall z \in \mathbb{Z} \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$$

$$\begin{array}{ll} 1 \ \forall z \in \mathbb{Z} \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z & \text{Th. 17.3.1.7} \\ 2 \ \forall z \in \mathbb{Z} \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z & \text{Th. 17.3.1.8} \\ 3 \ \forall z \in \mathbb{Z} \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z \wedge \forall z \in \mathbb{Z} \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z \wedge I(1, 2) \end{array}$$

Geordnete Nachfolgerstruktur

Th. 17.5.2.1(iii)

$$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}}) \wedge \forall z \in \mathbb{Z} \ (z < S_{\mathbb{Z}}(z))$$

$$\begin{array}{ll} 1 \ \text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}}) & \text{Th. 17.4.2.3} \\ 2 \ \forall z \in \mathbb{Z} \ (z < S_{\mathbb{Z}}(z)) & \text{Th. 17.4.2.4} \\ 3 \ \text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}}) \wedge \forall z \in \mathbb{Z} \ (z < S_{\mathbb{Z}}(z)) \wedge I(1, 2) \end{array}$$

Induktionsschema

Th. 17.5.2.1(iv)

$$P(0_{\mathbb{Z}}), \ \forall z \in \mathbb{Z} \ (P(z) \rightarrow P(S_{\mathbb{Z}}(z))), \\ \forall z \in \mathbb{Z} \ (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \forall z \in \mathbb{Z} \ P(z)$$

$$\begin{array}{ll} 1 \ 1 \ P(0_{\mathbb{Z}}) & A \\ 2 \ 2 \ \forall z \in \mathbb{Z} \ (P(z) \rightarrow P(S_{\mathbb{Z}}(z))) A & \\ 3 \ 3 \ \forall z \in \mathbb{Z} \ (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) A & \\ 1,2,3 \ 4 \ \forall z \in \mathbb{Z} \ P(z) & \text{Th. 17.5.1.1(1,2,3)} \end{array}$$

Schluss:

$$1 \ \mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}} \quad \text{Def. 17.2.2.1}$$

2	$0_{\mathbb{Z}} \in \mathbb{Z}$	$\wedge E1(\text{Th. 17.5.2.1}(i))$
3	$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	$\wedge E1(\wedge E2(\text{Th. 17.5.2.1}(i)))$
4	$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	$\wedge E2(\wedge E2(\text{Th. 17.5.2.1}(i)))$
5	$5 \ z \in \mathbb{Z}$	A
5	$6 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$	$\rightarrow E(5, \wedge E1(\text{Th. 17.5.2.1}(ii)))$
5	$7 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$	$\rightarrow E(5, \wedge E2(\text{Th. 17.5.2.1}(ii)))$
8	$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$	$\wedge E1(\text{Th. 17.5.2.1}(iii))$
5	$9 \ z < S_{\mathbb{Z}}(z)$	$\rightarrow E(5, \wedge E2(\text{Th. 17.5.2.1}(iii)))$
10	$10 \ P(0_{\mathbb{Z}})$	A
11	$11 \ \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z)))$	A
12	$12 \ \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z)))$	A
10,11,12	$13 \ \forall z \in \mathbb{Z} P(z)$	$\text{Th. 17.5.2.1}(iv)(10,11,12)$
14	$\text{ZPeano}(\mathbb{Z}, 0_{\mathbb{Z}}, S_{\mathbb{Z}}, P_{\mathbb{Z}}, \leq_{\mathbb{Z}})$	$\text{Def. 17.5.2.1}(2,3,4,6,7,8,9,13)$
15	$\mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}} \wedge \text{ZPeano}(\mathbb{Z}, 0_{\mathbb{Z}}, S_{\mathbb{Z}}, P_{\mathbb{Z}}, \leq_{\mathbb{Z}}) \wedge I(1, 14)$	

□

Axiom 17.5.2.1 (Null-Axiom der ganzen Zahlen).

$$0_{\mathbb{Z}} \in \mathbb{Z}$$

Herkunft aus der Quotientenkonstruktion: Th. 17.2.3.5.

Axiom 17.5.2.2 (Nachfolgerfunktion der ganzen Zahlen).

$$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

Herkunft aus der Quotientenkonstruktion: Th. 17.3.1.1.

Axiom 17.5.2.3 (Vorgängerfunktion der ganzen Zahlen).

$$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

Herkunft aus der Quotientenkonstruktion: Th. 17.3.1.2.

Axiom 17.5.2.4 (Vorgänger nach Nachfolger).

$$z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$$

Herkunft aus der Quotientenkonstruktion: Th. 17.3.1.7.

Axiom 17.5.2.5 (Nachfolger nach Vorgänger).

$$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$$

Herkunft aus der Quotientenkonstruktion: Th. 17.3.1.8.

Axiom 17.5.2.6 (Totale Ordnung der ganzen Zahlen).

$$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$$

Herkunft aus der Quotientenkonstruktion: Th. 17.4.2.3.

Axiom 17.5.2.7 (Strikte Nachfolgerorientierung).

$$z \in \mathbb{Z} \vdash z < S_{\mathbb{Z}}(z)$$

Herkunft aus der Quotientenkonstruktion: Th. 17.4.2.4.

Axiom 17.5.2.8 (Zweiseitiges Induktionsaxiom).

$$\begin{aligned} &P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))), \\ &\forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \forall z \in \mathbb{Z} P(z) \end{aligned}$$

Herkunft aus der Quotientenkonstruktion: Th. 17.5.1.1.